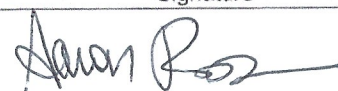
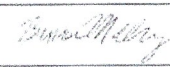


	Design Review		Doc No.: FM-0052
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Project No.	Product Name	Part No./ Rev	Review Phase	Review Date
00017	Hydrophilix 100% Silicone Foley Catheter	HDX-8000 / B HDX-8001 / B HDX-8002 / B	Phase 4 Design Verification & Validation	08/25/2020
Subject of Review				
#	Topic	Details	Reference Documents	
1	Action Items from Last Review (08/20/2020)	No action items	FILE-005990 rev 08/25/2020	
2	Label Content Review	The document was reviewed. The requirement is complete.	FILE-005989 rev 08/25/2020	
3	Biocompatibility	The test report was reviewed. The requirement is complete.	TR-0071 rev B	
4	Packaging & Sterile Barrier	The documents were reviewed. The requirement is complete.	TP-0058 rev A, TR-0074 rev A	
5	Product Performance	The documents were reviewed. The requirement is complete.	TP-0056 rev B, TR-0072 rev A	
6	Sterilization	The document was reviewed. The requirement is complete.	TR-0075 rev B	
7	Stability / Shelf Life	The documents were reviewed. The requirement is complete.	TP-0057 rev B, TR-0073 rev A	
8	Design FMEA (Risk)	The document was reviewed. The requirement is complete.	RM-0020 rev A	
9	Product Specification / Trace Matrix	No new specifications, nor changes required to the Product Requirements. Trace matrix Inputs, Outputs, Verification and Validation is complete.	PS-0006 rev B	
10	Project Plan	No changes required to the project plan. The plan is current.	PL-0013 rev B	
11	Regulatory Plan	No change required to the plan. The plan is current.	PL-0014 rev B	
12	Risk	No new risks discovered, nor changes required to the RMP or HA. The risk management plan is current. With no new risks identified the HA is complete.	RM-0018 rev A, RM-0019 rev A	
13	Review of Phase 4 Deliverables	Each document was reviewed and discussed. Project team in attendance has concluded that the deliverables are complete and satisfy the Design Control requirements in accordance with QP-007 rev E. It is agreed upon to close PHASE 4 and progress into PHASE 5.	Refer to #1-#2, QP-007 rev E.	
Action Items				
#	Description		Assigned To	Due Date
N/A	N/A		N/A	N/A

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Project No.	Product Name	Part No./ Rev	Review Phase	Review Date
00017	Hydrophilix 100% Silicone Foley Catheter	HDX-8000 / B HDX-8001 / B HDX-8002 / B	Phase 4 Design Verification & Validation	08/25/2020

Reviewers					
Functional Group	Required?	Name	Title	Signature	Date
Quality Assurance / Regulatory Affairs	☒	Aaron Rogers	Director of Regulatory and Quality		8/25/2020
R&D/ Engineering / Manufacturing	☒	Jonas Cochran	Managing Partner		8/25/2020
Marketing	☐				
Independent Reviewer	☒	Tom Cassou	Director of Business Development		8.25.2020
Other / Customer	☒	Brian McVerry	Chief Technology Officer		8.25.20



HYDROPHILIX

2-WAY 100% SILICONE FOLEY CATHETER

SIZE X Fr X ml

QUANTITY X

#005

R_Xonly

#006



#025



STERILE EO



#035

MR

LOT

XXXXXXXX

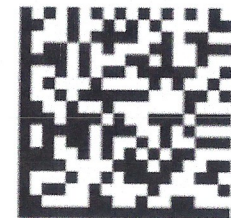
REF

XXXX



YYYY-MM-DD

#009



#010

(01)00851682007XXX

#012 (10)1234567

(17)YYMMDD

#004 Symbols are defined in Instruction for Use (IFU)

Store away from direct light exposure at room temperature

#084 Not made with natural rubber latex

Distributed by:

Hydrophilix, Inc. | 12100 Wilshire Blvd, Suite 800 | Los Angeles, CA 90025 | (949) 735 2614

HDX-4000 Rev.C

#001

#002

ARop 08/25/2020

#003 ENGLISH



HYDROPHILIX

#005 **R_x Only**

100% Silicone 2-Way Foley Catheter

Balloon Inflation Volume: 14 Fr / 18 Fr 10 ml, 16 Fr 5 ml

Please read all instructions before using this device

#084 Not made with natural rubber latex

#085 Foley Catheter is MR Safe.

#025 **Contraindications**
#021

This device is contraindicated in the presence of conditions which create unacceptable risk during catheterization.

#014 **Intended Use**
#021

Urological catheter is intended for drainage of the urinary tract. Catheterization is accomplished by inserting the catheter through the urethra and into the bladder. Intended population is adults and pediatrics.

#006 **Directions**

#016 **Installation & Inflation**

- #020
1. Prepare application site in a sterile fashion.
 2. Visually inspect the catheter for any signs of damage or deterioration prior to use.
 3. If a trial inflation is conducted, follow steps 7 & 8 on the sterile field then evacuate the sterile water with the syringe. Ensure the balloon is empty prior to insertion.
 4. Apply medical grade lubricant to distal end of the catheter.
 5. Gently advance the catheter through the urethra until distal tip is beyond the bladder neck.
 6. Confirm the position of the distal tip. The return of urine typically indicates the distal tip is in the bladder.
 7. Fill Luer-tipped syringe with sterile water and connect to the inflation valve.
 8. Fill the catheter balloon to the labeled volume. Do NOT exceed labeled volume.
 9. Slowly withdraw the catheter until resistance is felt. This is a sign the balloon is resting at the bladder neck.

Deflation & Removal

1. Gently connect Luer-tipped syringe to the inflation valve. Gently aspirate while allowing the pressure in the balloon to assist filling the syringe with water occupying the balloon. Remove and reconnect the syringe in the inflation valve if there is no or slow deflation. Rapid deflation could collapse the lumen and prevent adequate deflation of the balloon.
2. If during the removal process the balloon ruptures, ensure all fragments are carefully removed from patient.
3. Discard after use.

ARoz 08/25/2020



HYDROPHILIX

#021
#025

Precautions

1. Sterile technique shall be applied during installation and removal of the catheter.
2. It is recommended to be replaced after 7 days of use, however, shall not exceed 30 days.

#016
#017

#021
#025

Cautions

1. Federal law restricts the device to sale or by the order of a physician.
2. Never clamp the catheter. Use catheter plug or valve if required.

#005

#021
#025

Warnings

1. Single Use ONLY. Do NOT reuse. Reuse may cause infection, illness, injury or death.
2. Read the instructions carefully before using the device.
3. Do not use this device beyond the labeled expiration date.
4. Do not use the device if the packaging is punctured or damaged.
5. Do not force catheter as injury to the urethra can occur.
6. Do not use needle.
7. Do not exceed recommended capacities.

#021
#025

Potential Adverse Events

With all Foley Catheters, the following potential adverse events could occur:

1. Urinary tract infection
2. Perforation or laceration of the urethra
3. Perforation or laceration of the bladder
4. Abrasions

#007

#004

Symbol Description

REF

Catalog Number

LOT

Lot Number



Use by



Consult instructions for use



Single Use



Sterilized by Ethylene Oxide



Do not resterilize



Do not use if package is damaged

Rx Only

Prescription Use Only (US)



Foley Catheter is MR Safe

Distributed by
Hydrophilix, Inc.

12100 Wilshire Blvd, Suite 800
Los Angeles, CA 90025
(949) 735-2614

#001

#002

#022 HDX-4001 rev C

ARogm 08/25/2020

#003 ENGLISH

	Label Content Design Requirements	Doc No.: WI-0009
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5.0 Responsibilities

- 5.1 It is the responsibility of the individual that designs product labels and labeling to comply with this procedure.

6.0 General Requirements

- 6.1 Paper stock and printing materials shall be selected based on its ability to maintain legibility throughout the labeling life (e.g., processing, storage, handling, and distribution).
- 6.2 Adhesive, if applicable, shall be selected based on its ability to maintain adherence throughout the labeling life (e.g., processing, storage, handling, and distribution).
- 6.3 Additional labeling requirements maybe required based on product specific regulations or guidance documents.

7.0 Labeling Requirements

#	Reference	Requirement Description	Device / Label / IFU	Markets
✓ 0001	21CFR801.1 21CFR812.5	Device label shall include the name of manufacturer, packer, or distributor. <i>Abbreviations for "Incorporated", etc. may be used. Qualifying phrase to distinguish relationship "Manufactured for ____", etc. may be used.</i>	Label	US
✓ 0002	21CFR801.1 21CFR812.5	Device label shall include street address, city, state, and zip code of manufacturer, packer, or distributor. <i>Street address may be omitted if listed in current directory.</i>	Label	US
✓ 0003	21CFR801.15	The predominant language shall be English.	Label / IFU	US
✓ 0004	21CFR801.15	Use of symbols shall include either adjacent text in all language(s) or a glossary of symbols.	Label / IFU	US
✓ 0005	21CFR801.15 21CFR801.109 FDA (1150)	The symbol statement "Rx only" or "[rx] only" or the statement "Caution: Federal law restricts this device to sale by or on the order of a ____" (i.e. "physician", etc.) <i>Except for surgical instruments.</i> "Only" needs to immediately follow "Rx." However, the symbol statement "Rx only" does not necessarily need to be bracketed in quotation marks, and the word "only" may appear in upper or lower case letters, for example, Rx only, Rx Only, or Rx ONLY.	Label / IFU	US
✓ 0006	21CFR801.109	The label of the device includes the method of its application or use. <i>Except for surgical instruments.</i>	Label / IFU	US
✓ 0007	21CFR801.15	If a glossary of symbols is used, it shall contain: - Each symbol used in the labeling - Title and designation number of standard(s) - Title of symbol and reference number (if any) - The symbols meaning	IFU	US
✓ 0008 N/A	21CFR801.15	If any representation appears in a foreign language, then all labeling requirements shall also appear in that foreign language.	Label / IFU	US
✓ 0009	21CFR801.18	Dates shall be expressed in the format YYYY-MM-DD (i.e. 4-digit year, 2-digit month, 2-digit day, each separated by a hyphen).	Label / IFU	US

	Label Content Design Requirements		Doc No.: WI-0009
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#	Reference	Requirement Description	Device / Label / IFU	Markets
0010	21CFR801.20 MDR Annex I 23.2	Label shall include a unique device identifier (UDI).	Label	US
0011	21CFR801.40	UDI must be presented in two forms: - Plain-text - Automatic Identification and Data Capture (AIDC)	Label	US
0012	21CFR801.40	UDI includes the following information: - Device identifier, which is a fixed portion that identifies the specific version or model of a device and the labeler of the device - Production identifier, which is a variable portion that identifies one or more of the following: lot number, serial number, expiration date, manufacturing date	Label	US
0013 N/A	21CFR801.45	Device meant to be reprocessed must be directly marked with UDI in either or both: - Plain-text - Automatic Identification and Data Capture (AIDC) or alternative technology <i>Except when it interferes with safety and efficacy, or not technically feasible.</i>	Device	US
0014	21CFR801.5	Instruction for Use includes statements of all conditions, purposes or uses for which such device is intended, prescribed or recommended.	IFU	US
0015	21CFR801.5	Instruction for Use includes Quantity of dose for: - each intended use - persons of different ages and physical conditions	IFU	US
0016	21CFR801.5 21CFR801.109	Instruction for Use includes frequency of administration or application.	IFU	US
0017	21CFR801.5 21CFR801.109	Instruction for Use includes duration of administration or application.	IFU	US
0018	21CFR801.5 21CFR801.109	Instruction for Use includes time of administration or application in relation to meals, onset of symptoms or other time factors.	IFU	US
0019	21CFR801.5 21CFR801.109	Instruction for Use includes route or method of administration or application.	IFU	US
0020	21CFR801.5	Instruction for Use includes preparation for use.	IFU	US
0021	21CFR801.109	Instruction for Use includes indications for Use, Effects, Hazards, Contraindications, side effects, precautions.	IFU	US
0022	21CFR801.109	Instruction for Use includes date of issuance or latest revision.	IFU	US
0023 N/A	21CFR812.5	An Investigational Device Exemption (IDE) label includes the quantity of contents.	Label	US
0024 N/A	21CFR812.5	An IDE label includes "CAUTION--Investigational device. Limited by Federal (or United States) law to investigational use."	Label	US
0025	21CFR812.5	The label or other labeling shall describe all relevant contraindications, hazards, adverse effects, interfering substances or devices, warnings, and precautions.	Label / IFU	US
0026 N/A	21CFR812.5	The labeling of an investigational device shall not bear any statement that is false or misleading in any particular and shall not represent that the device is safe or effective for the purposes for which it is being investigated.	Label / IFU	US



Label Content Design Requirements

Doc No.: WI-0009

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#	Reference	Requirement Description	Device / Label / IFU	Markets
0027 N/A	21CFR812.5	An investigational device shipped solely for research on or with laboratory animals shall bear on its label the following statement: "CAUTION--Device for investigational use in laboratory animals or other tests that do not involve human subjects."	Label	US
0028	MDR Annex I 23.1	Information provided (medium, format, content, legibility, location) is appropriate for the device and healthcare practitioner.	Label / IFU	EU
0029	MDR Annex I 23.1	Where practical, information shall appear on the device itself.	Device	EU
0030	MDR Annex I 23.1	Information on device labeling shall take the form of symbols, where appropriate. In areas for which no standards exist, the symbols and colors shall be described in the documentation supplied with the device.	Label / IFU	EU
0031	MDR Annex I 23.1	Labels shall be provided in human-readable format and supplemented with machine-readable information.	Label	EU
0032	MDR Annex I 23.1	Each device shall be accompanied by Instructions for Use (IFU). <i>Except for class I and IIa devices that don't require information for safe use.</i>	IFU	EU
0033	MDR Annex I 23.1, 23.2, 23.3	Device name shall appear on the device label, sterile barrier packaging and IFU.	Label / IFU	EU
0034	MDR Annex I 23.1, 23.2, 23.3	Details needed to identify the device / device description shall appear on the device label, sterile barrier packaging.	Label	EU
0035	MDR Annex I 23.1, 23.2	Packaging contents shall appear on the device label.	Label	EU
0036	MDR Annex I 23.1, 23.2, 23.4	Intended for use statement shall appear on the device label.	Label	EU
0037	MDR Annex I 23.1, 23.2, 2.3, 23.4	Name and address of the manufacturer shall appear on the device label, sterile barrier packaging, IFU.	Label / IFU	EU
0038	MDR Annex I 23.2	Name and Address of authorized representative shall appear on the device label. <i>Except where the manufacturer resides within the EU.</i>	Label	EU
0039	MDR Annex I 23.2, 23.4	Where applicable, device label and IFU shall include an indication device contains: - Medicinal substance, human blood, plasma derivative - Tissues, cells or derivatives of human origin - Tissues, cells or derivatives of animal origin	Label / IFU	EU
0040	MDR Annex I 23.2, 23.4	Where applicable, device label and IFU shall include an indication device contains: - Substances which are carcinogenic, mutagenic or toxic to reproduction - Substances having endocrine-disrupting properties	Label / IFU	EU
0041	MDR Annex I 23.2	The label shall include the lot or serial number of the device along with the appropriate symbol.	Label	EU
0042	MDR Annex I 23.2, 23.3	The sterile barrier packaging shall include the date of manufacture (YYYY-MM).	Label	EU
0043	MDR Annex I 23.2, 23.3	The label and sterile barrier packaging shall include the expiration date (YYYY-MM), if applicable. Otherwise the label shall include the date of manufacture.	Label	EU
0044	MDR Annex I 23.2	The label shall include the UDI number.	Label	EU



Label Content Design Requirements

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#	Reference	Requirement Description	Device / Label / IFU	Markets
0045	MDR Annex I 23.2, 23.4	The label and IFU shall indicate any storage and / or handling conditions that apply.	Label / IFU	EU
0046	MDR Annex I 23.3	The sterile barrier packaging shall be indicated it is the sterile barrier.	Label	EU
0047	MDR Annex I 23.2, 23.3, 23.4	The label, sterile barrier packaging, and IFU shall indicate its method and state of sterility that applies, with the appropriate symbol.	Label / IFU	EU
0048	MDR Annex I 23.2	The label shall include any immediate warnings or precautions that apply.	Label	EU
0049	MDR Annex I 23.2, 23.4	The label and IFU shall indicate the device is for single use with the appropriate symbol, if that applies.	Label / IFU	EU
0050	MDR Annex I 23.2	The label shall indicate the device is a reprocessed single-use device, and the number of reprocessing cycles performed any limitations that apply.	Label	EU
0051	MDR Annex I 23.2	The label and sterile barrier packaging shall indicate the device is "custom-made" if applicable.	Label	EU
0052	MDR Annex I 23.2	The label shall indicate the device is a medical device.	Label	EU
0053	MDR Annex I 23.2, 23.3	The label and sterile barrier packaging shall indicate the device is "exclusively for clinical investigation" if applicable.	Label	EU
0054	MDR Annex I 23.2, 23.4	The label and IFU shall indicate the device is composed of / combination of substances intended to be introduced / applied.	Label / IFU	EU
0055	MDR Annex I 23.2	The label shall indicate the serial number for implantable devices, if applicable.	Label	EU
0056	MDR Annex I 23.3	The sterile barrier packaging shall include an instruction to check the IFU if the sterile packaging is damaged or unintentionally opened before use.	Label	EU
0057	MDR Annex I 23.4	Instructions for Use includes device's intended purpose with a clear specification of indications, contra-indications, the patient target group or groups, and of the intended users, as appropriate.	IFU	EU
0058	MDR Annex I 23.4	Instructions for Use includes where applicable, a specification of the clinical benefits to be expected.	IFU	EU
0059	MDR Annex I 23.4	Instructions for Use includes where applicable, links to the summary of safety and clinical performance.	IFU	EU
0060	MDR Annex I 23.1, 23.4	Instructions for Use includes the performance characteristics of the device.	IFU	EU
0061	MDR Annex I 23.4	Instructions for Use includes where applicable, information allowing the healthcare professional to verify if the device is suitable and select the corresponding software and accessories.	IFU	EU
0062	MDR Annex I 23.4	Instructions for Use includes where applicable, information allowing the healthcare professional to verify if the device is suitable and select the corresponding software and accessories.	IFU	EU
0063	MDR Annex I 23.1, 23.4	Instructions for Use includes any residual risks, limitations, contra-indications, precautions, warnings and any undesirable side-effects, including information to be conveyed to the patient in this regard.	IFU	EU

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#	Reference	Requirement Description	Device / Label / IFU	Markets
0064	MDR Annex I 23.4	Instructions for Use includes specifications the user requires to use the device appropriately (e.g. degree of accuracy).	IFU	EU
0065	MDR Annex I 23.4	Instructions for Use includes details of any preparatory treatment or handling of the device before it is ready for use or during its use, such as sterilisation, final assembly, calibration, etc., including the levels of disinfection required to ensure patient safety and all available methods for achieving those levels of disinfection.	IFU	EU
0066	MDR Annex I 23.4	Instructions for Use includes any requirements for special facilities, or special training, or particular qualifications of the device user and/or other persons.	IFU	EU
0067	MDR Annex I 23.4	Instructions for Use includes the information needed to verify whether the device is properly installed and is ready to perform safely and as intended.	IFU	EU
0068	MDR Annex I 23.4	Instructions for Use includes instructions in the event of the sterile packaging being damaged or unintentionally opened before use, if supplied sterile.	IFU	EU
0069	MDR Annex I 23.4	Instructions for Use includes appropriate instructions for sterilization, if applicable.	IFU	EU
0070	MDR Annex I 23.4	Instructions for Use includes if the device is reusable, information on the appropriate processes for allowing reuse, including cleaning, disinfection, packaging and, where appropriate, the validated method of re-sterilization.	IFU	EU
0071	MDR Annex I 23.4	Instructions for Use includes an indication, if appropriate, that a device can be reused only if it is reconditioned under the responsibility of the manufacturer to comply with the general safety and performance requirements.	IFU	EU
0072	MDR Annex I 23.4	Instructions for Use includes information on known characteristics and technical factors known to the manufacturer that could pose a risk if the device were to be re-used. Applied to single use devices.	IFU	EU
0073	MDR Annex I 23.4	Instructions for Use includes information to identify other devices and equipment in order to obtain a safe combination, and / or any known restrictions. Applies to devices intended for use together with other devices and/or general-purpose equipment.	IFU	EU
0074	MDR Annex I 23.4	Instructions for Use includes detailed information as to the nature, type and where appropriate, the intensity and distribution of the emitted radiation and the means of protecting the patient, user, or other person from unintended radiation during use of the device. Applies to devices that emit radiation for medical purposes.	IFU	EU
0075	MDR Annex I 23.4	Instructions for Use includes information that allows the user and/or patient to be informed of any warnings, precautions, contra- indications, measures to be taken and limitations of use regarding the device.	IFU	EU
0076	MDR Annex I 23.4	Instructions for Use includes the overall qualitative and quantitative information on the materials and substances to which patients can be exposed with implantable devices.	IFU	EU
0077	MDR Annex I 23.4	Instructions for Use includes warnings or precautions to be taken in order to facilitate the safe disposal of the device, its accessories and the consumables used with it, if any.	IFU	EU

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#	Reference	Requirement Description	Device / Label / IFU	Markets
0078	MDR Annex I 23.4	Instructions for Use includes the circumstances in which the user should consult a healthcare professional, if intended for use by lay persons.	IFU	EU
0079	MDR Annex I 23.4	Instructions for Use includes date of issue of the instructions for use or, if they have been revised, date of issue and identifier of the latest revision of the instructions for use.	IFU	EU
0080	MDR Annex I 23.4	Instructions for Use includes notice to the user and/or patient that any serious incident that has occurred in relation to the device should be reported to the manufacturer and the competent authority of the Member State in which the user and/or patient is established.	IFU	EU
0081	MDR Annex I 23.4	Instructions for Use includes information to be supplied to the patient with an implanted device.	IFU	EU
0082	MDR Annex I 23.4	Instructions for Use includes, for devices that incorporate electronic programmable systems, including software, or software that are devices in themselves, minimum requirements concerning hardware, IT networks characteristics and IT security measures, including protection against unauthorized access, necessary to run the software as intended.	IFU	EU
0083	MDR Article 20#5, Annex V	CE conformity marking shall consist of the initials "CE" followed by the identification number of the notified body responsible for the device's conformity assessment. The various components of the CE marking must have substantially the same vertical dimension, which may not be less than 5 mm.	Label / IFU	EU
0084	FDA (1768)	If a manufacturer elects to include a statement in medical product labeling indicating that natural rubber latex or synthetic derivatives of natural rubber latex were not used as materials in the manufacture of their medical product and container, FDA recommends the use of the statement "Not made with natural rubber latex." If this statement is made without any qualification, it would apply to the medical product, its container, and any packaging.	Label / IFU	US
0085	ASTM F2503	For devices which are brought into the magnetic resonance (MR) environment, permanently mark the item as MR Safe, MR Conditional, or MR Unsafe using the icons with color or black and white options. Refer to ASTM F2503 TABLE 1 and TABLE 2 for details.	Device / Label / IFU	US

8.0 Records

8.1 Label Content documents will be processed per QP-002. Refer to QP-003 for storage location and retention requirements.

9.0 Flowchart – N/A

10.0 Change History

Revision	Change Description
A	Initial release. This procedure will replace procedure QP-7.09
B	Update header, references, and format of requirements (table).

	Test Report, Biocompatibility, HDX Foley Catheter		Doc No.: TR-0071
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1.0 Purpose

- 1.1 The purpose of the report is to summarize the results of the biocompatibility evaluations performed for the HDX Foley Catheter.

2.0 Scope

- 2.1 The biocompatibility test results reported applies to the HDX Foley Catheters. The affected part numbers are: HDX-8000, HDX-8001, HDX-8002.
- 2.2 Test units were representative of the finished good and sterilized at the contract sterilization facility, Steris, located at 43425 Business Park Dr, Temecula, CA 92590.
- 2.3 Biocompatibility testing was performed at the qualified contract test laboratory, Nelson Labs, 6280 S. Redwood Road, Salt Lake City, UT 84123 USA
- 2.4 Biocompatibility testing was performed at the qualified contract test laboratory, WUXI Apptec, 2540 Executive Drive, St. Paul, MN 55120 USA

3.0 References

- 3.1 ISO 10993-1:2009 – Biological evaluation of medical device – Part 1: Evaluation and testing within a risk management process

4.0 Background

- 4.1 The HDX Foley Catheter is constructed in medical grade silicone. The working length is considered patient contacting, which is the outer surface of the shaft, balloon, and distal tip. The outer surface of the patient contacting region is treated with UV curing hydrophilic polymer solution. Based on the clinical application, the device is categorized as a surface-contacting/ mucosal membrane with a prolong contact duration (> 24 hours to 30 days) per ISO 10993-1. The biocompatibility test program for device category is:

- Cytotoxicity
- Sensitization
- Irritation or Intracutaneous Reactivity

In addition, the following biocompatibility test were conducted:

- Systemic Injection
- Material mediated pyrogen
- Intramuscular Implantation
- Subacute Intraperitoneal systemic toxicity
- Sub chronic Intravenous systemic toxicity

5.0 Sampling Requirements

- 5.1 Sampling requirements for each biocompatibility evaluation are summarized in the following tables and were indicated in each laboratory protocol attached to the laboratory reports indicated in §11.0.

Table 5.1 – Sampling Requirements				
Evaluation Description	Study No. / Protocol No.	Sample Quantity		
		14Fr	16Fr	18Fr
Cytotoxicity: MEM elution, 72 hr incubation., L929, 24 hr extended	1169158-S01 / STP-0032 rev 10	1	1	1
Sensitization: Magnusson-Kligman Method, 2 extracts	1169160-S01 P17-1773-02A	6	6	6
Irritation: Intracutaneous Toxicity (ISO), 2 extracts	1169161-S01 P17-1771-01A	2	2	2
Systemic Toxicity: Systemic Injection (ISO), 2 extracts	1169162-S01 P17-1778-01A	2	2	2
Systemic Toxicity: Material Mediated Pyrogen, each	1169163-S01 P17-1777-01A	1	1	1
Intramuscular Implantation	D00041453 902300-25	0	0	1
Sub acute systemic toxicity	D00041453 800592-10	0	0	1
Sub chronic systemic toxicity	D00041453 800593-10	0	1	0

6.0 Material and Equipment

- 6.1 Materials and equipment required for each biocompatibility evaluation are indicated in laboratory protocol attached to the laboratory reports indicated in §11.0.

6.2 Test Article

6.2.1 The articles consisted of the following:

- HYDPLX-14Fr-ENG001 (HDX-8000)
- HYDPLX-16Fr-ENG001 (HDX-8001)
- HYDPLX-18Fr-ENG001 (HDX-8002)
- HYDPLX-16Fr-ENG007 (HDX-8001)
- HYDPLX-18Fr-ENG006 (HDX-8002)

7.0 Acceptance Criteria

- 7.1 Acceptance criteria for each biocompatibility evaluation are per ISO 10993 standards and indicated in laboratory protocol attached to the laboratory reports indicated in §11.0.

8.0 Deviations

- 8.1 No deviations were reported in the laboratory reports.

9.0 Results & Discussion

	Test Report, Biocompatibility, HDX Foley Catheter		Doc No.: TR-0071
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- 9.1 Each catheter size was treated with UV curing hydrophilic polymer solution and then packaged into a chevron Tyvek pouch.
- 9.2 Test articles were terminally sterilized via Ethylene Oxide (EO) by Steris (Temecula, CA). Refer to Attachment 11.1 for additional details.
- 9.3 Test articles were shipped to Nelson Laboratories (Salt Lake City, UT) and Toxikon (Bedford, MA) for the biocompatibility evaluations. The evaluations were conducted under GLP conditions.
- 9.4 For subacute/sub chronic systemic toxicity and implantation studies, test articles were shipped to WUXI Apptec (St. Paul; MN). The evaluations were conducted under GLP conditions.
- 9.5 The following are the summarized conclusions from each biocompatibility evaluation.

Table 9.4 – Biocompatibility Testing Summary		
Evaluation Description	Result	Meets Acceptance Criteria
Cytotoxicity: MEM elution, 72 hr incubation., L929, 24 hr extended	The materials tested are considered non-toxic. All test method acceptance criteria met. Refer to Attachment §11.5	Yes
Sensitization: Magnusson-Kligman Method, 2 extracts	The materials tested elicited no sensitization reaction. The test article is classified as a non-sensitizer. Refer to Attachment §11.6	Yes
Irritation: Intracutaneous Toxicity (ISO), 2 extracts	The test articles sites did not show a significantly greater biological reaction than the sites with the control article. The test article meets the requirement of the ISO 10993-10 guidelines. Refer to Attachment §11.7	Yes
Systemic Toxicity: Systemic Injection (ISO), 2 extracts	The test article did not induce a significantly greater biological reaction than the control extracts following a single dose to Albino Swiss mice. The test article meets the requirement of the ISO 10993-11 guidelines. Refer to Attachment §11.8	Yes
Systemic Toxicity: Material Mediated Pyrogen, each	The test article is considered non-pyrogenic and meets the requirements of the Pyrogen Test, ISO 10993-11 guidelines. Refer to Attachment §11.9	Yes
Intramuscular Implantation	Clinical Observations: All animals survived to the scheduled study endpoint. No abnormal clinical signs were noted for any of the animals during the course of the study. Macroscopic Observations: At explant, the right lymph node for 59383 was difficult to visualize. In addition, one implant was found outside of the muscle for rabbit 59385. Otherwise, the explant observations were normal. Microscopic Observations: The average of the test article site scores (10.7) minus the average of the control site scores (11.2) equals the irritant ranking score (-0.5). A negative value is considered zero. Based on the observations of the study pathologist an irritant ranking score of 0.0 was calculated indicating the test article was a non-irritant as compared to the USP high density polyethylene reference standard. The complete histopathology report is attached to this report. Conclusion: Based on the gross and the histopathologic evaluations, the test article was considered a non-irritant.	Yes

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Subacute IP Toxicity	All animals in the test and control groups survived for the duration of the 28-day study. No abnormal clinical signs were noted in any of the animals during daily observations. All animals gained weight over the course of the study. There was no mean body weight loss in any group over the course of the study. There were no statistically significant differences (all p values were greater than 0.05) between the test and control rats body weights at any recorded time period in the study. No abnormal findings were noted at necropsy for test and control animals. The pathology analysis concluded there were no adverse findings that could be attributed to the test article. Although statistically significant differences were present in Hematology, Coagulation, Clinical Chemistry and Organ Weight data; those findings were not considered to be indicative of a pattern of toxicity. Conclusion: Under the conditions of this study the sesame oil extract of the test article is considered negative for signs of systemic toxicity due to leachable components.	Yes
Subchronic IV Toxicity	All animals in the test and control groups survived for the duration of the 28-day study. No abnormal clinical signs were noted in any of the animals during daily observations. All animals gained weight over the course of the study. There was no mean body weight loss in any group over the course of the study. There were no statistically significant differences (all p values were greater than 0.05) between the test and control animal body weights at any recorded time period in the study. No abnormal findings were noted at necropsy for the test animals. The pathology analysis concluded there were no adverse findings that could be attributed to the test article. Although statistically significant differences were present in Hematology and Clinical Chemistry data; those findings were not considered to be indicative of a pattern of toxicity. Conclusion: Under the conditions of this study the normal saline extract of the test article is considered negative for signs of systemic toxicity due to leachable components.	Yes

10.0 Conclusion

- 10.1 The HDX Foley Catheter met all biological evaluation requirements indicated in the report, which is in-line with a surface-contacting/ mucosal membrane with a prolong contact duration (> 24 hours to 30 days).

11.0 Attachments

- 11.1 DHR- HYDPLX-14Fr-ENG001 (HDX-8000)
- 11.2 DHR- HYDPLX-16Fr-ENG001 (HDX-8001)
- 11.3 DHR- HYDPLX-18Fr-ENG001 (HDX-8002)
- 11.4 DHR- HYDPLX-16Fr-ENG007 (HDX-8001)
- 11.5 DHR- HYDPLX-18Fr-ENG006 (HDX-8002)
- 11.6 Steris Sterilization Record, Cycle No. 21015

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- 11.7 Steris Sterilization Record, Cycle No. 22569
- 11.8 Nelson Labs Study No 1169158-S01 Cytotoxicity Report
- 11.9 Nelson Labs Study No 1169160-S01 Sensitization Report
- 11.10 Nelson Labs Study No 1169161-S01 Intracutaneous Report
- 11.11 Nelson Labs Study No 1169162-S01 Systemic Injection Report
- 11.12 Nelson Labs Study No 1169163-S01 Material Mediated Pyrogen Report
- 11.13 WUXI Report No 239627 ISO Intramuscular Implantation Report
- 11.14 WUXI Report No 240814 ISO Subacute IP Toxicity Report
- 11.15 WUXI Report No 241793 ISO Subchronic IV Toxicity Report

12.0 Change History

Revision	Change Description
A	Initial release
B	Add Intramuscular Implantation, Subacute & Subchronic systemic toxicity studies

Pathway Innovation LLC.	Transportation Challenge, HDX Foley Catheter	Doc No.: TP-0058
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1.0 Purpose

- 1.1 This protocol provides the requirements for conducting the transportation challenge for the HDX Foley Catheter.

2.0 Scope

- 2.1 The applies to corrugated and sterile barrier packaging that protects the HDX Foley Catheter and its sterility during transportation.

3.0 References

- 3.1 QP-002 – Document Control
- 3.2 QP-030 – Statistical Analysis
- 3.3 QP-031 – ETO Sterilization Program
- 3.4 ASTM 4169-16 – Standard Practice for Performance Testing for Shipping Containers and System
- 3.5 ASTM D4332-14 – Standard Practice for Conditioning Containers, Packages, or Packaging Components for Testing
- 3.6 ASTM F1886/F1886M-16 – Standard Test Method for Determining Integrity of Seals for Medical Packaging by Visual Inspection
- 3.7 TM-0005 – Test Method, Gross Leak Detection
 - 3.7.1 ASTM F2096-11 – Standard Test Method for Detecting Gross Leaks in Packaging by Internal Pressurization (Bubble Test)
- 3.8 TM-0003 – Test Method, Seal Integrity (Seal Strength)
 - 3.8.1 ASTM F88/F88M-15 – Standard Test Method for Seal Strength of Flexible Barrier Materials

4.0 Responsibilities

- 4.1 It is the responsibility of the individual performing this procedure to comply with all requirements and instructions indicated.

5.0 General Requirements

- 5.1 Traceability of components and assemblies utilized in this protocol are required. The minimum traceability requirements to be maintained is the part number, revision, and lot number. Records of traceability will be incorporated in the final report.
- 5.2 Personnel that is conducting the evaluation activities shall be properly trained to the protocol and records of the training will incorporated in the final report.
- 5.3 Any deviation to the approved protocol will be documented in the final report.

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5.4 All equipment used will be calibrated (if applicable). Equipment asset numbers and calibration due dates will be documented on the data collection sheets.

5.5 Results of evaluation will be documented on the appropriate data collection sheet.

5.5.1 Attribute results will be recorded with a “P” for Pass” and “F” for “Fail”.

6.0 Sampling Requirements

6.1 Refer to the following table for the sample size for each size per test method and the sequence of evaluation:

Table 6.1 – Sample Requirements Table				
Sequence	Test Method	Sample Size/ French Size	Destructive (D)/ Non-Destructive (ND)	Attribute/ Variable
		18 Fr		
A.1	Climatic Conditioning	Six (6) 10-Packs (60 Units)	ND	A
A.2	Simulated Distribution	Six (6) 10-Packs (60 Units)	ND	A
A.3	Visual Inspection	58 Units	ND	A
A.4	Bubble Emission	58 Units	ND	A
A.5	Seal Strength	30 Units	D	V
A.6	Device Visual Inspection	58 Units	ND	A

6.2 18 Fr catheters were selected because they are considered worst-case for this evaluation since they have the largest mass of the size selection (i.e., 14, 16 Fr). The packaging is identical for all product sizes.

7.0 Material and Equipment

7.1 Climatic and Simulated Distribution Conditioning

7.1.1 Equipment will be described in Contract Test Facilities Test Report.

7.2 Gross Leak Detection (Bubble Emission)

7.2.1 Refer to TM-0005

7.3 Seal Strength

7.3.1 Refer to TM-0003.

8.0 Equipment Setup

8.1 Climatic and Simulated Distribution Conditioning

8.1.1 Equipment will be described in Contract Test Facilities Test Report.

8.2 Gross Leak Detection (Bubble Emission)

8.2.1 Refer to TM-0005.

8.3 Seal Strength

8.3.1 Refer to TM-0003.

9.0 Acceptance Criteria

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9.1 Corrugated Packaging Visual Inspection

- 9.1.1 There cannot be any separation of the adhesive bond holding the seam together or damage that could impact the devices incorporated.

9.2 Sterile Barrier Packaging Performance

9.2.1 Visual Inspection

- 9.2.1.1 The seals shall be complete and uniform and no channels or voids can be observed through the entire width of the seal.

9.2.2 Gross Leak Detection (Bubble Emission)

- 9.2.2.1 There shall be no gross leaks when subjected to the determined internal test pressure and test duration.

9.2.3 Seal Strength

- 9.2.3.1 The sterile barrier seal shall have a tensile strength value greater than the minimum specification of one pound (1 lb.).

- 9.2.3.2 k-factor analysis will be conducted which the minimum confidence interval shall be $\geq$ the minimum specification when a k-factor constant of 2.220 (95% confidence/95% reliability) is applied in the following equation:

$$CI = \bar{X} \pm (k - factor \bullet \sigma)$$

Where,

CI = Confidence Interval

$\bar{X}$ = mean

k-factor = constant (2.220)

σ = Standard Deviation

9.3 Device Visual Inspection

- 9.3.1 There cannot be any damage to device where the safety and efficacy is compromised.

10.0 Procedure

- 10.1 Product will be terminally sterilized via Ethylene Oxide (EO) per QP-031. Evidence of the processing will be incorporated in the final report.

- 10.2 The six (6) 10-packs will be inserted in an appropriately sized corrugated box that minimally meets the bursting test specification of 200 lbs./in². Packaging material (e.g., bubble wrap) will be used to fill voids.

- 10.3 The evaluation units will be provided to an approved contract test facility to conduct climatic conditioning per ASTM D4332 then simulated distribution per D4169. Contract test facility will provide records of processing, which will be attach to this report.

- 10.4 Inspect the corrugated packaging from a distance of 12 to 18 inches. Record results on the Data Collection Sheets.

- 10.5 Sterile Barrier Packaging Performance

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10.5.1 Visual inspect the seal at a distance of 12 to 18 inches. Inspect the entire sealed area of the package for completeness and uniformity. Identify and record on the Data Collection Sheet any part of the seal where channels appear.

10.5.2 Bubble emission will be conducted per TM-0005. Record results on the Data Collection Sheets.

10.5.3 Seal strength will be conducted per TM-0003. Record results on the Data Collection Sheets.

10.6 Inspect the devices from a distance of 12 to 18 inches. Record results on the Data Collection Sheets.

11.0 Records

11.1 The records generated from this evaluation will be incorporated into a report, which is be retained per QP-002.

12.0 Attachment

12.1 Date Collection Sheet, Transportation Challenge

13.0 Change History

Revision	Change Description
A	Initial Release

Pathway Innovation LLC.	Transportation Challenge Adoption, HDX Foley Catheter	Doc No.: TR-0074
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1.0 Purpose

- 1.1 The purpose of this report is to document the assessment of the transportation challenge testing requirement for HDX Foley Catheter.

2.0 Scope

- 2.1 This applies to corrugated and sterile barrier packaging that protects the HDX Foley Catheter and its sterility during transportation.

3.0 References

- 3.1 TP-0058 – Test Protocol, Transportation Challenge, HDX Foley Catheter
- 3.2 TR-0068 – Test Report, Transportation Challenge, PSM Foley Catheter

4.0 Acceptance Criteria

- 4.1 Corrugated Packaging Visual Inspection
 - 4.1.1 There cannot be any separation of the adhesive bond holding the seam together or damage that could impact the devices incorporated.
- 4.2 Sterile Barrier Packaging Performance
 - 4.2.1 Visual Inspection
 - 4.2.1.1 The seals shall be complete and uniform and no channels or voids can be observed through the entire width of the seal.
 - 4.2.2 Gross Leak Detection (Bubble Emission)
 - 4.2.2.1 There shall be no gross leaks when subjected to the determined internal test pressure and test duration.
 - 4.2.3 Seal Strength
 - 4.2.3.1 The sterile barrier seal shall have a tensile strength value greater than the minimum specification of one pound (1 lb.).
- 4.3 Device Visual Inspection
 - 4.3.1 There cannot be any damage to device where the safety and efficacy is compromised.

5.0 Deviations

- 5.1 Rather than perform the prescribed testing, an assessment / paper adoption was performed.

6.0 Results & Discussion

- 6.1 The PSM Foley Catheter sterile barrier and packaging configuration was qualified to protect the catheter and its sterility per TR-0068. The activities conducted in the transportation challenge were: 1) Sterile Barrier Performance Evaluation per TP-0016, 2) Simulated Distribution evaluations per ASTM D4169 and 3) device visual inspection per TP-0016. All acceptance criteria were met.

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6.2 A comparison of the device configuration of the HDX Foley Catheter and PSM Foley Catheter are summarized in the table below.

Table 6.2 – Device Configuration Assessment		
Device Configuration Description	HDX Foley Catheter	PSM Foley Catheter
Foley Catheter Length (in)	16.5	16.5
Foley Catheter Width (in)	1.3	2.0
Foley Catheter Size (FR)	14, 16, 18	14, 16, 18
Foley Catheter Balloon Volume (ml)	10 – 30	10 – 30
Foley Catheter Lumens / Branches	2	3
Material	Silicone	Silicone

6.2.1 The difference in device configuration is that the HDX Foley Catheter lacks the 3rd lumen branch. This does not create more of a challenge than the PSM Foley Catheter. The overall footprint of the device is smaller than the PSM Foley Catheter. Therefore, the device does not create more of challenge to the packaging.

6.3 A comparison of the packaging configuration of the HDX Foley Catheter and PSM Foley Catheter are summarized in the table below.

Table 6.3 – Packaging Configuration Assessment		
Package Configuration Description	HDX Foley Catheter	PSM Foley Catheter
Pouch, Chevron, 3.5 X 21, Amcor TLP-058	Yes	Yes
Bag, 2" W x 18" L x 2 Mil Thk, Polyethylene	Yes	Yes
White Carton, 10-Pack, Inner, 20.75 L x 4.12 W x 2.37 H	Yes	Yes
Corrugate Box, 22 in. L x 14 in. W x 12 in. H, 275 LB Test, Double Wall	Yes	Yes
Units / White Carton	10	10
Units / Corrugate Box	100	100

6.3.1 There are no differences in the packaging configuration. The same components are used on both configurations.

6.4 A comparison of the packaging weight of HDX Foley Catheter and PSM Foley Catheter are summarized in the table below.

Table 6.3 – Packaging Weight Assessment		
Package Property Description	HDX Foley Catheter	PSM Foley Catheter
Weight of 100-Pack Corrugate Box	9.06 LB (145 oz)	9.46 LB (151 oz)
Volume of 100-Pack Corrugate Box	3696 in ³	3696 in ³
Density of 100-Pack Corrugate Box	.04 oz/in ³	.04 oz/in ³

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6.4.1 The weight difference of the 100-pack of HDX Foley Catheters is 6 oz. less than the PSM Foley Catheters, or about 4%. Less weight would create less of a challenge to the packaging configuration.

6.5 Per §6.2, the HDX Foley Catheter configuration does not create a greater challenge to the packaging as the PSM Foley Catheter. Since it has been demonstrated through package performance testing and simulated distribution testing that the sterile barrier and packaging configuration was sufficient to protect the device (refer to TR-0068), it is highly likely that the packaging will continue to perform with the HDX Foley Catheter.

6.6 Per §6.3, the HDX Foley Catheter incorporates the same sterile barrier packaging configuration, the same inner carton, and outer distribution box as the PSM Foley Catheter. Since it has been demonstrated through package performance testing and simulated distribution testing that the sterile barrier and packaging configuration was sufficient to protect the device (refer to TR-0068), it is highly likely that the packaging will continue to perform with the HDX Foley Catheter.

6.7 Per §6.4, the HDX Foley Catheter is 6 oz. less than the PSM Foley Catheter. At the distribution box level, the weight difference is negligible and does not create a greater challenge to the packaging configuration. Based on this information, the HDX packaging configuration is equivalent to the packaging of the PSM Foley Catheter. Since it has been demonstrated through package performance testing and simulated distribution testing that the sterile barrier and packaging configuration was sufficient to protect the device (refer to TR-0068), it is highly likely that the packaging will continue to perform with the HDX Foley Catheter.

7.0 Conclusion

7.1 Based on this assessment, the packaging configuration of the HDX Foley Catheter is an equivalent challenge to the packaging configuration to protect the product and its sterile barrier after simulated distribution testing, when compared to the PSM Foley Catheter.

7.2 Adopt the sterile barrier performance testing and simulated distribution testing results of the PSM Foley Catheter for the HDX Foley Catheter.

8.0 Attachments

8.1 None

9.0 Change History

Revision	Change Description
A	Initial release

Pathway Innovation LLC.	Design Verification, HDX Foley Catheter	Doc No.: TP-0056
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1.0 Purpose

- 1.1 This protocol provides the requirements for conducting design verification performance testing for the HDX Foley Catheter.

2.0 Scope

- 2.1 Applies to the HDX Foley Catheter.

3.0 References

- 3.1 QP-002 – Document Control
3.2 QP-030 – Statistical Techniques
3.3 ASTM F623-99 – Standard Performance Specification for Foley Catheter
3.4 ISO 80369-1:2010 – Small-bore connectors for liquids and gas in healthcare applications – Part 1: General requirements

4.0 Responsibilities

- 4.1 It is the responsibility of the individual conducting the evaluation to comply with requirements indicated in this procedure.

5.0 General Requirements

- 5.1 Personnel conducting evaluation activities shall be properly trained to the protocol.
5.2 Any deviation to the approved protocol will be documented in the final report.
5.3 All equipment used will be calibrated (if applicable). Equipment asset numbers and calibration due dates will be documented on the data collection sheets.
5.4 Results will be documented on the appropriate data collection sheet.
5.4.1 Attribute results will be recorded with a “P” for Pass” and “F” for “Fail”.

6.0 Sampling Requirements

- 6.1 Refer to the following table for the sample size for each size per test method and the sequence of evaluation:

Table 6.1 – Sample Requirements Table						
Sequence	Test Method	Sample Size/ French Size			Destructive (D)/ Non-Destructive (ND)	Attribute/ Variable
		14 Fr	16 Fr	18 Fr		
A	Balloon Size and Shaft Size	5	5	5	ND	V
B.1	Flow Rate through Drainage Lumen	22	22	22	ND	V
B.2	Balloon Volume Maintenance	22	22	22	ND	A
B.3	Deflation Reliability (Failure to Deflate)	22	22	22	ND	A
C	Balloon Integrity (Resistance to Rupture)	22	22	22	D	A

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Table 6.1 – Sample Requirements Table						
Sequence	Test Method	Sample Size/ French Size			Destructive (D)/ Non-Destructive (ND)	Attribute/ Variable
		14 Fr	16 Fr	18 Fr		
D.1	Verification Non-Interconnectable Connector	22	22	22	ND	A
D.2	Inflated Balloon Response to Traction	22	22	22	D	A

- 6.2 With the exception of the balloon size volume for the indicated sizes, the connectors, material, and construction are similar. Based on likeness between sizes, the sample sizes between the sizes are pulled together to achieve a confidence/reliability level of 95%/95% with no failures observed.
- 6.3 Evaluations that are dependent on balloon size volume will not be pulled together and will achieve independently a confidence/reliability level of 95%/90%.

7.0 Material and Equipment

7.1 Flow Rate through Drainage Lumen

- 7.1.1 Water Reservoir with Stopcock, capable of maintaining 20 ± 1.0 cm (7.9 ± 0.4 in.) of water (196 ± 10 kPa) above the tip of the catheter connection throughout the test
- 7.1.2 Graduated Cylinder, 200 cm³ or equivalent, Calibrated
- 7.1.3 Syringe, 10cc or equivalent
- 7.1.4 Distilled or Deionized Water
- 7.1.5 Stopwatch or Timer, Calibrated.

7.2 Balloon Integrity (Resistance to Rupture)

- 7.2.1 Tank/Container, Corrosion-Resistant
- 7.2.2 Heat Source, Waterproof
- 7.2.3 Thermometer, Calibrated
- 7.2.4 Syringe, 10cc or equivalent
- 7.2.5 Distilled or Deionized Water
- 7.2.6 Artificial Urine, Fresh - The pH is adjusted to 5.5 to 7.0 range by adding a 1 normal (N) solution of NH₄OH or NH₄Cl to a 1000-mL volumetric flask. The ingredients are the following:
- NaCl = 6.17 gm
 - NaH₂PO₄ = 4.59 gm
 - Na₃ Citrate = 0.944 gm
 - MgSO₄ = 0.463 gm
 - Na₂SO₄ = 2.41 gm
 - KCl = 4.75 gm
 - CaCl₂ = 0.64 gm
 - Na₂ Oxalate = 0.043 gm
 - Distilled water = 1000 gm

7.3 Inflated Balloon Response to Traction

Pathway Innovation LLC.	Design Verification, HDX Foley Catheter	Doc No.: TP-0056
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7.3.1 Funnel, 28 Fr barrel outlet, 6cm funnel barrel, 6cm “V” portion opening

7.3.2 Weight, 0.45 kg (1 lb.), Calibrated

7.3.3 Ring Stand

7.3.4 Stopwatch or Timer, calibrated

7.4 Balloon Volume Maintenance

7.4.1 Syringe, 10cc or equivalent

7.4.2 Distilled or Deionized Water

7.4.3 Methylene Blue Crystal Solution or Equivalent - Prepare 1 g of methylene crystals and dilute in 2000 cm³ of water

7.4.4 Stopwatch or Timer, calibrated

7.5 Balloon Size and Shaft Size

7.5.1 Caliper, Calibrated

7.5.2 Ruler, Steel

7.5.3 Scissor or equivalent

7.6 Deflation Reliability (Failure to Deflate)

7.6.1 Tank/Container, Corrosion-Resistant, 10-gallon capacity

7.6.2 Heat Source, Waterproof

7.6.3 Thermometer, Calibrated

7.6.4 Syringe, 10cc or equivalent

7.6.5 Scissor or equivalent

7.6.6 Distilled or Deionized Water

7.6.7 Caliper, Calibrated

7.7 Verification Non-Interconnectable Connector

7.7.1 Small-Bore Connector

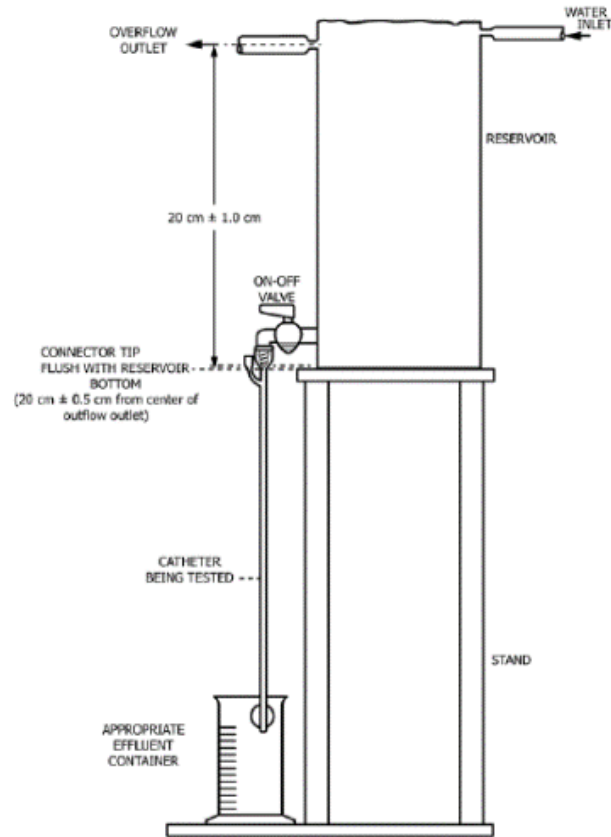
7.7.2 Mechanical Tensile Stand w/ Force Gage, Calibrated

8.0 Equipment Setup

8.1 Flow Rate through Drainage Lumen

8.1.1 Refer to the following illustration for the equipment set-up.

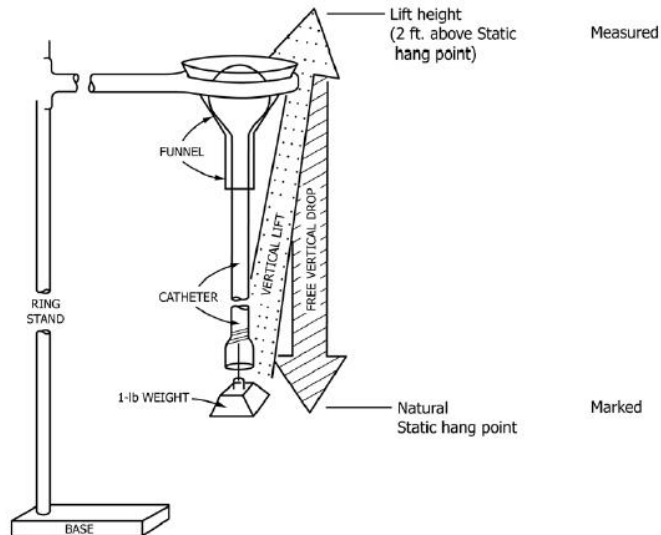
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8.1.2 The flow rate is adjusted through the water inlet to a rate sufficient to maintain flow through the overflow outlet while each catheter is tested. A head pressure of 20 ± 1.0 cm of water (196 ± 10 kPa) above the tank bottom shall be maintained throughout the test to approximate actual physiologic conditions. The overflow outlet should not be covered by water.

8.2 Inflated Balloon Response to Traction

8.2.1 Refer to the following illustration for the equipment set-up.



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9.0 Acceptance Criteria

9.1 Flow Rate through Drainage Lumen

9.1.1 Minimum average flow rate of 100 cm³/minutes

Note: For 14 – 24 French Catheters

9.2 Balloon Integrity (Resistance to Rupture)

9.2.1 The inflation balloon must be inflated easily with distilled or deionized water to its indicated labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml) without showing any evidence of breakage throughout the 7-day test period.

9.3 Inflated Balloon Response to Traction

9.3.1 The entire balloon of catheters shall not pass into or through the funnel barrel and shall not rupture.

9.4 Balloon Volume Maintenance

9.4.1 The catheter shall maintain its labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml) for 24-hour period.

9.5 Balloon Size and Shaft Size

9.5.1 The requirements on size and tolerances on diameter shall be met for the catheter tip (± 1 Fr), the balloon (+4 Fr), and the shaft, for 10 in. distal to the balloon (± 1 Fr) for sterile catheter directly removed from the sterile barrier packaging.

Note: The balloon section may wrinkle but shall not tear or distort, and the catheter shaft or tip may offer resistance but if distortion or stretching occurs it is considered a failure.

9.6 Deflation Reliability (Failure to Deflate)

9.6.1 The balloon shall deflate to within 4 French sizes of the label French size within 15 minutes or be otherwise manipulated to effect drainage after the 7-day immersion period at its labeled fill volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml).

9.7 Verification Non-Interconnectable Connector

9.7.1 The catheter connection shall not appear to provide a secure connection when forcefully assembled to any surface of the components and shall easily disengage from each of the Small-Bore Connector.

10.0 Procedure

10.1 Flow Rate through Drainage Lumen

10.1.1 Inflate the balloon of the catheter with distilled or deionized water to its labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml).

10.1.2 Connect the proximal end of the drainage lumen to the reservoir stopcock. The proximal end of the catheter connection at the stopcock should be level with the bottom of the tank ± 1 cm and it should deliver fluid at 20 ± 1 cm (196 $\pm$ 10 kPa) head pressure at that junction. Refer to illustration in §8.1.2

10.1.3 Establish the inflow and outflow equilibrium before test measurements. Refer to §8.1.2

10.1.4 Record the amount of water (in cm³) flowing through the drainage lumen in 1 minute.

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- 10.1.5 Test each catheter 3 times and calculate the average flow rate for each catheter. Record results in the Data Collection Sheets.

10.2 Balloon Integrity (Resistance to Rupture)

- 10.2.1 Fill the tanks with artificial urine and raise the temperature to $37.8 \pm 3^{\circ}\text{C}$.
- 10.2.2 Inflate the balloons of the catheter with distilled or deionized water to its labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml). .
- 10.2.3 Fully submerge at least the entire balloon of the catheter in the tank.
- 10.2.4 After 7 days, inspect the catheters for rupture, leaks, or loss of volume in the balloons. Record results on the Data Collection Sheets.

10.3 Inflated Balloon Response to Traction

- 10.3.1 Pass the proximal tip of the catheter through the funnel barrel, advancing the balloon past the funnel inlet. Refer to illustration in §8.2.1.
- 10.3.2 Inflate the balloon with 5 cm³ of distilled or deionized water.
- 10.3.3 Pull the balloon back slowly until it fits snugly against the funnel inlet.
- 10.3.4 Hang a 0.45-kg (1-lb) weight from the distal end of the catheter.
- 10.3.5 Static Load Test
 - 10.3.5.1 Observe the catheter balloon for 2 minutes.
 - 10.3.5.2 Record result on the Data Collection Sheets.
- 10.3.6 Impact Load Test
 - 10.3.6.1 There should be no obstruction to the free fall of the weight from a height of 305 mm (1 ft) above the base of the ring stand during its natural suspension.
 - 10.3.6.2 Lift the weight 0.6 m (2 ft) from the point of its natural suspension and then release.
 - 10.3.6.3 Observe the balloon position in the funnel inlet after the drop.
 - 10.3.6.4 Record results on the Data Collection Sheets.

10.4 Balloon Volume Maintenance

- 10.4.1 Inflate the balloon of the catheter with the methylene blue solution to its labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml).
- 10.4.2 Place on a surface suitable for detection of color leakage for a 24-hour period. Cover or otherwise protect the catheters from light or ozone for the duration of the test.
- 10.4.3 After the 24-hour period has elapsed, observe the surfaces for any discoloration caused by content leakage from the catheter balloon-inflation system. Record results on the Data Collection Sheets.

10.5 Balloon Size and Shaft Size

- 10.5.1 Without lubrication, measuring the catheter tip, balloon diameter, and 254 mm (10") section of the shaft with calipers. Make sure that there is no force applied with the calipers when taken measurement.
- 10.5.2 Record results on the Data Collection Sheets.

10.6 Deflation Reliability (Failure to Deflate)

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- 10.6.1 Inflate the balloons of the catheters with distilled or deionized water to its labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml).
- 10.6.2 Place the catheters in an immersion bath for a period of 7 days at $37.8 \pm 3^{\circ}\text{C}$ ($100 \pm 5^{\circ}\text{F}$). Cover them to ensure complete immersion and to prevent evaporation and light damage.
- 10.6.3 After 7 days immersion, remove the catheters, place on clean surface and empty in accordance with the manufacturer's recommended technique or by use of the appropriate syringe or by the "cut-off" technique.
 - 10.6.3.1 Any catheter that fails to deflate using the syringe shall be cut off approximately 12.7 mm (1/2 in.) proximal to the bifurcation of the inflation lumen and drainage lumen.
- 10.6.4 Post deflation, the shape of the balloon portion of the catheter shall be no larger than 4 Fr sizes than the labeled French size using a calibrated French gage. A force of no greater than 4.5 N (1 lb.) may be used to pull the catheter through.

10.7 Verification Non-Interconnectable Connector

- 10.7.1 Prior to assembly condition the catheter connector at $23 \pm 2^{\circ}\text{C}$ and a relative humidity of $50\% \pm 5\%$ for no less than 1 hour.
- 10.7.2 Assemble the small-bore connector to the catheter connector by applying an axial force at a rate of approximately 10 N/second. The axial force should not visibly damage either connector and cannot exceed 70 N and a torque not exceeding 0.12 Nm to a limit of no more than 90° . Rotate threaded connector in a clockwise manner. Hold the maximum assembly force for no less than 10 seconds.
- 10.7.3 After 10 seconds of engagement, without activation of any latch or disengagement mechanism, apply an axial force of separation to the assembled connectors to a maximum 0.02 N (2 g). Verify the assembled connectors disengage.
- 10.7.4 If applicable, for threaded connectors, repeat steps 10.7.1 – 10.7.3 by assembling the connectors in a counterclockwise manner in step 10.7.2.
- 10.7.5 Repeat steps 10.7.1 – 10.7.4 for every potential assembly surface.
- 10.7.6 Record results in the Data Collection Sheets.

11.0 Records

- 11.1 The records generated from this evaluation will be incorporated into a report, which is be retained per QP-002.

12.0 Attachments

- 12.1 Data Collection Sheet

13.0 Change History

Revision	Change Description
A	Initial release
B	Revise for balloon fill volume of 5ml / 10ml

Pathway Innovation LLC.	Design Verification Report, HDX Foley Catheter	Doc No.: TR-0072
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1.0 Purpose

- 1.1 The purpose of this report is to document the results of the design verification evaluations associated with HDX Foley Catheter per protocol TP-0056.

2.0 Scope

- 2.1 This report applies to the 14, 16, 18 Fr HDX Foley Catheter at time equals zero (T=0).

3.0 References

- 3.1 TP-0056 – Test Protocol, Design Verification, HDX Foley Catheter

4.0 Acceptance Criteria

4.1 Flow Rate through Drainage Lumen

- 4.1.1 Minimum average flow rate of 100 cm³/minutes

Note: For 14 – 24 French Catheters

4.2 Balloon Integrity (Resistance to Rupture)

- 4.2.1 The inflation balloon must be inflated easily with distilled or deionized water to its indicated labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5 ml) without showing any evidence of breakage throughout the 7-day test period.

4.3 Inflated Balloon Response to Traction

- 4.3.1 The entire balloon of catheters shall not pass into or through the funnel barrel and shall not rupture.

4.4 Balloon Volume Maintenance

- 4.4.1 The catheter shall maintain its labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5 ml) for 24-hour period.

4.5 Balloon Size and Shaft Size

- 4.5.1 The requirements on size and tolerances on diameter shall be met for the proximal catheter tip (± 1 Fr), the balloon (+4 Fr), and the shaft, for 10 in. distal to the balloon (± 1 Fr) for sterile catheter directly removed from the sterile barrier packaging.

Note: The balloon section may wrinkle but shall not tear or distort, and the catheter shaft or tip may offer resistance but if distortion or stretching occurs it is considered a failure.

4.6 Deflation Reliability (Failure to Deflate)

- 4.6.1 The balloon shall deflate to within 4 French sizes of the label French size within 15 minutes or be otherwise manipulated to effect drainage after the 7-day immersion period at its labeled fill volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5 ml).

4.7 Verification Non-Interconnectable Connector

- 4.7.1 The catheter connection shall not appear to provide a secure connection when forcefully assembled to any surface of the components and shall easily disengage from each of the Small-Bore Connector.

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5.0 Deviations

5.1 There were no deviations reported during the evaluations.

6.0 Results & Discussion

6.1 All test measurement equipment used in this evaluation was calibrated and documented on the appropriate data collection sheet.

6.2 The data collection sheets provide product identification of the sterile catheters that were evaluated as part of the study.

6.3 Traceability Information

Part No.	Lot No.	Qty	Sterilization Load No.
HDX-8000	HYDPLX-14FR-ENG-003	50	21668
HDX-8001	HYDPLX-16FR-ENG-004	50	21668
HDX-8002	HYDPLX-18FR-ENG-005	50	21668

6.4 Flow Rate through Drainage Lumen

6.4.1 Twenty-two (22) units from each size was evaluated per test method indicated in TP-0056. The flow rate was measured on each catheter three (3) times then averaged. The results are indicated in the following table:

Table 6.4.1 – Statistical Summary, Flow Rate through Drainage Lumen			
	14Fr	16Fr	18Fr
Min	241.96	465.16	747.67
Max	295.63	499.54	807.12
Mean	280.08	482.84	776.90
StDev	12.38	9.57	17.62
Spec	≥100 cm ³ /min	≥100 cm ³ /min	≥100 cm ³ /min
Result	Pass	Pass	Pass

All catheter sizes evaluated resulted in a flow rate through the drainage lumen greater than 100 cm³/min. Acceptance criteria was met. Refer to attached data collection sheets for additional information.

6.5 Balloon Integrity (Resistance to Rupture)

6.5.1 Twenty-two (22) units from each size was evaluated per test method indicated in TP-0056. The catheters were filled according to their labeled fill volumes (i.e., 14/18 Fr = 10ml, 16 Fr = 5 ml). All units evaluated were not observed with leaks or breakage for the test duration. Acceptance criteria was met. Refer to attached data collection sheets for additional information.

6.6 Inflated Balloon Response to Traction

6.6.1 Twenty-two (22) units from each size was evaluated per test method indicated in TP-0056. A fill volume of 5ml was evaluated since it is worst case for this evaluation. All units evaluated did not rupture or pass into the or through the funnel barrel in the static or impact load evaluation. Acceptance criteria was met. Refer to attached data collection sheets for additional information.

6.7 Balloon Volume Maintenance

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- 6.7.1 Twenty-two (22) units from each size was evaluated per test method indicated in TP-0056. The catheters were filled according to their labeled fill volumes (i.e., 14/18 Fr = 10ml, 16 Fr = 5 ml). All units evaluated maintained the delivered fill volume for the test duration. Acceptance criteria was met. Refer to attached data collection sheets for additional information.

6.8 Balloon Size and Shaft Size

- 6.8.1 Five (5) units from each size was measured at the catheter tip, balloon, and shaft (10" distal to the balloon). The results are indicated in the following table:

Table 6.8.1 – Statistical Summary, Balloon Size and Shaft Size (inches)									
	14Fr (0.184)			16Fr (0.210)			18Fr (0.236)		
	Tip	Balloon	Shaft	Tip	Balloon	Shaft	Tip	Balloon	Shaft
Min	0.175	0.213	0.176	0.198	0.240	0.201	0.227	0.267	0.225
Max	0.182	0.233	0.182	0.205	0.254	0.205	0.232	0.280	0.232
Mean	0.177	0.222	0.179	0.201	0.245	0.203	0.229	0.274	0.229
StDev	0.003	0.008	0.003	0.003	0.005	0.001	0.002	0.005	0.003
Spec	± 1Fr (0.013)	+ 4Fr (0.052)	± 1Fr (0.013)	± 1Fr (0.013)	+ 4Fr (0.052)	± 1Fr (0.013)	± 1Fr (0.013)	+ 4Fr (0.052)	± 1Fr (0.013)
Result	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass

All catheter sizes evaluated were within the specified tolerance as it relates to the labeled size. Acceptance criteria was met. Refer to attached data collection sheets for additional information.

6.9 Deflation Reliability (Failure to Deflate)

- 6.9.1 Twenty-two (22) units from each size was evaluated per test method indicated in TP-0056. The catheters were filled according to their labeled fill volumes (i.e., 14/18 Fr = 10ml, 16 Fr = 5 ml). Refer to attached data collection sheets for additional information.
- 6.9.2 The results for post-deflation measurements are indicated in the following table:

Table 6.4.1 – Statistical Summary, Deflation Reliability (inches)			
	14Fr (0.184)	16Fr (0.219)	18Fr (0.236)
Fill Volume	10ml	5ml	10ml
Min	.206	.231	.252
Max	.220	.247	.266
Mean	.213	.240	.260
StDev	.004	.004	.003
Spec	+ 4Fr (0.052)	+ 4Fr (0.052)	+ 4Fr (0.052)
Result	Pass	Pass	Pass

None of the catheters evaluated were observed with a rupture or leak and they all deflated with no issues. The post-deflation measurements were within the specified 4Fr from the labeled size. Acceptance criteria was met. Refer to attached data collection sheets for additional information.

6.10 Verification Non-Interconnectable Connector

- 6.10.1 Twenty-two (22) units from each size was evaluated per test method indicated in TP-0056. The small-bore connector was observed being easily assembled to the catheters evaluated and easily disconnected with no issues. Acceptance criteria was met. Refer to attached data collection sheets for additional information.

7.0 Conclusion

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7.1 The 14, 16, 18 Fr HDX Foley Catheter met the acceptance criteria for functional, and leakage indicated in §4.0.

8.0 Attachments

- 8.1 Data Collection Sheet, 14Fr (Dated: 7/12/2019 – 7/27/2019)
- 8.2 Data Collection Sheet, 16Fr (Dated: 7/15/2019 – 7/27/2019)
- 8.3 Data Collection Sheet, 18Fr (Dated: 7/15/2019 – 7/27/2019)
- 8.4 DHR-8000_A_LN_HYDPLX-14FR-ENG-003
- 8.5 DHR-8001_A_LN_HYDPLX-16FR-ENG-004
- 8.6 DHR-8002_A_LN_HYDPLX-18FR-ENG-005
- 8.7 STERIS Sterilization Process Record Load 21668
- 8.8 Training Record – Megan Williams

9.0 Change History

Revision	Change Description
A	Initial release

Pathway Innovation LLC.	Sterilization Product Adoption, HDX 2-Way Foley Catheter	Doc No.: TR-0075
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1.0 Purpose

- 1.1 The purpose of the report is to provide the sterilization product adoption analysis for the HDX 2-Way Foley Catheter (Subject Device) that will allow it to be part of the sterilization product family for the validated EO cycle.

2.0 Scope

- 2.1 This applies to the Subject Device and the validated sterilization cycle (PWI-7374) that is located at Steris in Temecula, CA. The current product validated for the sterilization cycle is the PSM 3-Way Foley Catheter (Validated Device).

3.0 References

- 3.1 ISO 11135:2014 – Sterilization of health-care products — Ethylene oxide — Requirements for the development, validation and routine control of a sterilization process for medical devices
- 3.2 TIR28:2016 – Product adoption and process equivalency for ethylene oxide sterilization
- 3.3 TR-0067 – Report, Sterilization Validation, PSM Foley Catheter

4.0 Product Adoption Assessment

- 4.1 The following assessment was conducted with guidance to TIR28.

4.1.1 Nature and sources of raw materials

The Subject Device is made of the same material as the Validated Device, which is medical grade silicone, however the Subject Device is coated with a hydrophilic coating.

4.1.2 Size of Product

The Subject Device and the Validation Device are identical in length (16.5 Inches), size offering (14, 16, 18Fr), and balloon inflation volume (10-30ml). The Subject Device differs in that it does not incorporate an irrigation lumen while the Validated Device does. The irrigation lumen adds more internal/external surface area, weight, and density to the Validated Device. Refer to the following illustration of the Subject Device and Validated Device:

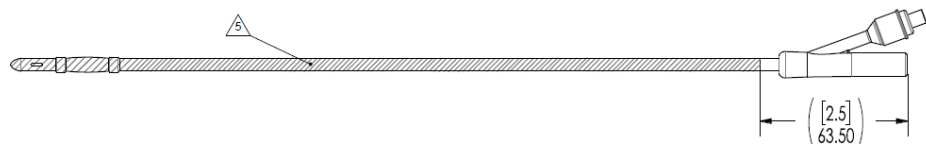


Figure 4.1.2a – Subject Device

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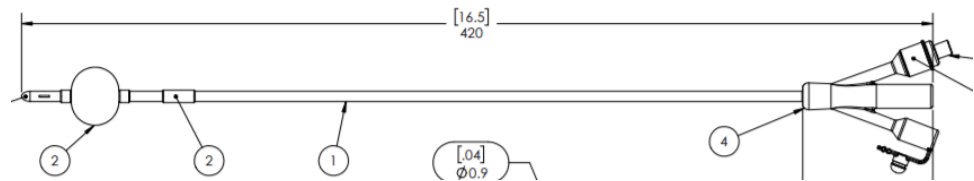


Figure 4.1.2b – Validated Device

4.1.3 Number of Components

4.1.3.1 Product

Subject Device and Validated Device is identical in its unit and carton quantities. One (1) device is packaged in the sterile barrier and ten (10) packaged devices are inserted in the carton.

4.1.3.2 Packaging

The sterile barrier and carton packaging are identical between the Subject Device and the Validated Device. The sterile barrier is a Tyvek/poly pouch.

4.1.4 Complexity of Product

The Subject Device is less complex than the Validated Device due to not having an irrigation lumen.

4.1.5 Manufacturing Processes

The manufacturing process for the Foley catheter is the same process for the Subject Device and the Validated Device. The raw silicone is heated to a liquid state and injected into a mold. The Subject Device is further processed in applying the hydrophilic coating.

4.1.6 Manufacturing Environment

The Subject Device is manufactured in the same environment as the Validated Device, which is a cleanroom.

4.1.7 Bioburden

The test laboratory was able to recover 54.5% from the Subject Device via exhaustive recovery method. The estimated bioburden was 471.8 CFU. Refer to the following table of bioburden summary results:

Table 4.1.7 – Bioburden Summary Results, Subject Device				
Type	Aerobic	Anaerobic	Fungal	Spores
Average	247.7 CFU	<43.3 CFU	<9.4 CFU	<205.8 CFU

Refer to the attached test report (Attachment 6.1) for additional details. The estimated bioburden is less than 1000 CFUs. The External Process Challenge Device (EPCD) has spore population of greater than 10^6 CFUs. The EPCD is at minimum 2000 times greater than the Subject Device's estimated bioburden.

4.1.8 EO Residual

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EO residuals evaluation was conducted with the Subject Device where units were subjected to the validated routine EO cycle and removed after six (6) hours heated aeration, which is the minimum validated aeration time that resulted in acceptable EO/ECH residual levels for a prolonged exposure (24 hours to 30 days) device. The results are indicated in the following table:

Table 4.1.8 - EO/ECH Residual Results		
	EO	ECH
Results	0.122 mg/device	0.091 mg/device
Specification	4 mg/device	9 mg mg/device
Pass/Fail	Pass	Pass

As indicated in the table, the EO/ECH residuals were below the maximum specification limit. Refer to Attachment 6.2 for additional detail.

5.0 Conclusion

- 5.1 Based on this analysis, the Subject Device can be adopted into the sterilization product family for the validated sterilization cycle. The size and complexity of the Validated Device is more challenging than the Subject Device. The Subject Device's EO residuals are expected to be equivalent or less than the Validated Device due to its lesser complexity, material similarity, and identical packaging configuration. The estimated bioburden of the Subject Device is significantly less than the EPCD that is used in the sterility release.

6.0 Attachments

- 6.1 Nelson Report 1192052-S01 – Bioburden Final Report (Dated: 6/12/2019)
- 6.2 Steris EO/ECH Residual Report (Dated: 7/26/2019).

7.0 Change History

Revision	Change Description
A	Initial Release
B	Added EO/ECH residual results.

Pathway Innovation LLC.	Stability Protocol, HDX Foley Catheter	Doc No.: TP-0057
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1.0 Purpose

- 1.1 This protocol provides the requirements for conducting both accelerated aging and real-time conditioning as well it identifies the evaluation requirements for the HDX Foley Catheter at each indicated interval.

2.0 Scope

- 2.1 The applies to the one (1) and three (3) year accelerated and real-time intervals for the HDX Foley Catheter.

3.0 References

- 3.1 QP-002 – Document Control
- 3.2 QP-030 – Statistical Analysis
- 3.3 QP-031 – ETO Sterilization Program
- 3.4 TP-0056 – Protocol, Design Verification, HDX Foley Catheter
- 3.5 PS-0006 – Product Specification & Trace Matrix, HDX Foley Catheter
- 3.6 ASTM F623-99 – Standard Performance Specification for Foley Catheter
- 3.7 WI-0011 – Work Instruction, Accelerated Aging Conditioning

4.0 Responsibilities

- 4.1 It is the responsibility of the individual performing this procedure to comply with all requirements and instructions indicated.

5.0 General Requirements

- 5.1 Traceability of components and assemblies utilized in this protocol are required. The minimum traceability requirements to be maintained is the part number, revision, and lot number. Records of traceability will be incorporated in the final report.
- 5.2 Personnel that is conducting the evaluation activities shall be properly trained to the protocol and records of the training will incorporated in the final report.
- 5.3 Any deviation to the approved protocol will be documented in the final report.
- 5.4 All equipment used will be calibrated (if applicable). Equipment asset numbers and calibration due dates will be documented on the data collection sheets.
- 5.5 Results of evaluation will be documented on the appropriate data collection sheet.
 - 5.5.1 Attribute results will be recorded with a “P” for Pass” and “F” for “Fail”.

6.0 Sampling Requirements

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- 6.1 Refer to the following table for the sample size for each size per test method and the sequence of evaluation:

Table 6.1 – Sample Requirements Table						
Sequence	Test Method	Sample Size/ French Size			Destructive (D)/ Non-Destructive (ND)	Attribute/ Variable
		14 Fr	16 Fr	18 Fr		
A.1	Device Performance Balloon Volume Maintenance	22	22	22	ND	A
A.2	Device Performance Balloon Integrity (Resistance to Rupture)	22	22	22	ND/D	A
A.3	Device Performance Deflation Reliability (Failure to Deflate)	22	22	22	ND/D	A
B.1	Device Performance Inflated Balloon Response to Traction	22	22	22	ND/D	A

- 6.1 With the exception of the balloon size volume for the indicated sizes, the connectors, material, and construction are similar. Based on likeness between sizes, the sample sizes between the sizes are pulled together to achieve a confidence/reliability level of 95%/95% with no failures observed.
- 6.2 Evaluations that are dependent on balloon size volume will not be pulled together and will achieve independently a confidence/reliability level of 95%/90%.
- 6.3 The test methods indicated in table 6.1 are associated with performance requirements that have potential impact as it relates the aging of the device. Other performance requirements (e.g., Balloon Size and Shaft Size, Flow Rate) are related the dimensional construction, which will not be impacted to product aging thus will not be evaluated during stability intervals.

7.0 Material and Equipment

7.1 Device Performance

7.1.1 Balloon Volume Maintenance

7.1.1.1 Syringe, 10cc or equivalent

7.1.1.2 Distilled or Deionized Water

7.1.1.3 Methylene Blue Crystal Solution or Equivalent - Prepare 1 g of methylene crystals and dilute in 2000 cm³ of water

7.1.1.4 Stopwatch or Timer, calibrated

7.1.2 Balloon Integrity (Resistance to Rupture)

7.1.2.1 Tank/Container, Corrosion-Resistant

7.1.2.2 Heat Source, Waterproof

7.1.2.3 Thermometer, Calibrated

7.1.2.4 Syringe, 10cc or equivalent

7.1.2.5 Distilled or Deionized Water

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7.1.2.6 Artificial Urine, Fresh - The pH is adjusted to 5.5 to 7.0 range by adding a 1 normal (N) solution of NH_4OH or NH_4Cl to a 1000-mL volumetric flask. The ingredients are the following:

- NaCl = 6.17 gm
- NaH_2PO_4 = 4.59 gm
- Na_3 Citrate = 0.944 gm
- MgSO_4 = 0.463 gm
- Na_2SO_4 = 2.41 gm
- KCl = 4.75 gm
- CaCl_2 = 0.64 gm
- Na_2 Oxalate = 0.043 gm
- Distilled water = 1000 gm

7.1.3 Deflation Reliability (Failure to Deflate)

7.1.3.1 Tank/Container, Corrosion-Resistant, 10-gallon capacity

7.1.3.2 Heat Source, Waterproof

7.1.3.3 Thermometer, Calibrated

7.1.3.4 Syringe, 10cc or equivalent

7.1.3.5 Scissor or equivalent

7.1.3.6 Distilled or Deionized Water

7.1.4 Inflated Balloon Response to Traction

7.1.4.1 Funnel, 28 Fr barrel outlet, 6cm funnel barrel, 6cm "V" portion opening

7.1.4.2 Weight, 0.45 kg (1 lb.), Calibrated

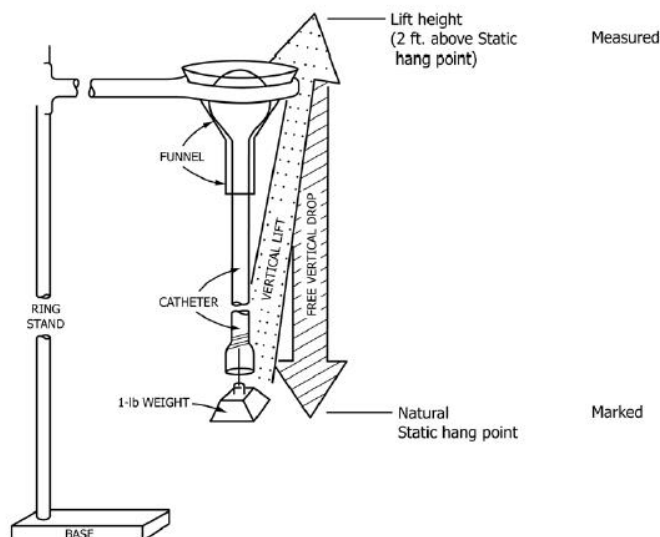
7.1.4.3 Ring Stand

7.1.4.4 Stopwatch or Timer, calibrated

8.0 Equipment Setup

8.1 Inflated Balloon Response to Traction

8.1.1 Refer to the following illustration for the equipment set-up.



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9.0 Acceptance Criteria

9.1 Device Performance

9.1.1 Balloon Volume Maintenance

- 9.1.1.1 The catheter shall maintain its labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml) for a 24-hour period.

9.1.2 Balloon Integrity (Resistance to Rupture)

- 9.1.2.1 The inflation balloon must be inflated easily with distilled or deionized water to its indicated labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml) without showing any evidence of breakage throughout the 7-day test period.

9.1.3 Deflation Reliability (Failure to Deflate)

- 9.1.3.1 The balloon shall deflate to within 4 French sizes of the label French size within 15 minutes or be otherwise manipulated to effect drainage after the 7-day immersion period at its labeled fill volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml)..

9.1.4 Inflated Balloon Response to Traction

- 9.1.4.1 The entire balloon of catheters shall not pass into or through the funnel barrel and shall not rupture.

10.0 Procedure

- 10.1 Product will be terminally sterilized via Ethylene Oxide (EO) per QP-031. Evidence of the processing will be incorporated in the final report.

10.2 Device Aging

- 10.2.1 The shelf-life periods that be evaluated are the following:

- One (1) year
- Three (3) years

10.2.2 Accelerated

- 10.2.2.1 The environment chamber will be set to 65°C, which the accelerated aging period for each shelf-life period is the following:

- One (1) years = 18 days
- Three (3) years = 55 Days

Note: Time period was determined per Arrhenius equation ($T = \text{Desired Shelf Life} / Q_{10}^Y$) indicated in WI-0011, which

$Q_{10} = 2.0$ [Acceleration Factor (Coefficient of Aging)]

$Y = 4.3$ [Elevated storage temperature (65°C) - Ambient storage temperature(22°C)/10]

- 10.2.2.2 Once the duration indicated is achieved then the product will be pull from the environmental chamber and evaluated to this procedure. *Note: A combination of real-time aging and accelerated aging can be combine to suffice the time duration.*

- 10.2.2.3 Record to accelerated aging will be documented and included in report.

10.2.3 Real-Time

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- 10.2.3.1 Product allocated to real-time aging will be identified as such on the product packaging with the part number, revision, and lot number. The date of each time intervals will be documented on the product packaging as well. This product will be retained in the designated location, which will be stored in ambient temperature (e.g., office environment) and no humidity controls. Once the duration indicated is achieved then the product will be evaluated to this procedure.

10.3 Device Performance

10.3.1 Balloon Volume Maintenance

- 10.3.1.1 Inflate the balloon of the catheter with the methylene blue solution to the its labeled fill volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml).
- 10.3.1.2 Place on a surface suitable for detection of color leakage for a 24-hour period. Cover or otherwise protect the catheters from light or ozone for the duration of the test.
- 10.3.1.3 After the 24-hour period has elapsed, observe the surfaces for any discoloration caused by content leakage from the catheter balloon-inflation system. Record results on the Data Collection Sheets.

10.3.2 Balloon Integrity (Resistance to Rupture)

- 10.3.2.1 Fill the tanks with artificial urine and raise the temperature to $37.8 \pm 3^{\circ}\text{C}$.
- 10.3.2.2 Inflate the balloons of the catheter with distilled or deionized water to its labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml).
- 10.3.2.3 Fully submerge at least the entire balloon of the catheter in the tank.
- 10.3.2.4 After 7 days, inspect the catheters for rupture, leaks, or loss of volume in the balloons. Record results on the Data Collection Sheets.

10.3.3 Deflation Reliability (Failure to Deflate)

- 10.3.3.1 Inflate the balloons of the catheters with distilled or deionized water to its labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml).
- 10.3.3.2 Place the catheters in an immersion bath for a period of 7 days at $37.8 \pm 3^{\circ}\text{C}$ ($100 \pm 5^{\circ}\text{F}$). Cover them to ensure complete immersion and to prevent evaporation and light damage.
- 10.3.3.3 After 7 days immersion, remove the catheters, place on clean surface and empty in accordance with the manufacturer's recommended technique or by use of the appropriate syringe or by the "cut-off" technique.
- 10.3.3.3.1 Any catheter that fails to deflate using the syringe shall be cut off approximately 12.7 mm (1/2 in.) proximal to the bifurcation of the inflation lumen and drainage lumen.
- 10.3.3.4 Post deflation, the shape of the balloon portion of the catheter shall be no larger than 4 Fr sizes than the labeled French size using a calibrated French gage. A force of no greater than 4.5 N (1 lb.) may be used to pull the catheter through.

10.3.4 Inflated Balloon Response to Traction

- 10.3.4.1 Pass the proximal tip of the catheter through the funnel barrel, advancing the balloon past the funnel inlet. Refer to illustration in §8.2.1.
- 10.3.4.2 Inflate the balloon with 5 cm³ of distilled or deionized water.

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10.3.4.3 Pull the balloon back slowly until it fits snugly against the funnel inlet.

10.3.4.4 Hang a 0.45-kg (1-lb) weight from the distal end of the catheter.

10.3.4.5 Static Load Test

10.3.4.5.1 Observe the catheter balloon for 2 minutes.

10.3.4.5.2 Record result on the Data Collection Sheets.

10.3.4.6 Impact Load Test

10.3.4.6.1 There should be no obstruction to the free fall of the weight from a height of 305 mm (1 ft) above the base of the ring stand during its natural suspension.

10.3.4.6.2 Lift the weight 0.6 m (2 ft) from the point of its natural suspension and then release.

10.3.4.6.3 Observe the balloon position in the funnel inlet after the drop.

10.3.4.6.4 Record results on the Data Collection Sheets.

11.0 Records

11.1 The records generated from this evaluation will be incorporated into a report, which is be retained per QP-002.

12.0 Attachment

12.1 Data Collection Sheet, Device Performance

13.0 Change History

Revision	Change Description
A	Initial Release
B	Revise for labeled volume of 5ml or 10ml

Pathway Innovation LLC.	Report, Stability, 1-Year Accelerated, HDX Foley Catheter	Doc No.: TR-0073
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1.0 Purpose

- 1.1 The purpose of this report is to document the results for evaluations associated to the stability protocol for the HDX Foley Catheter per TP-0057.

2.0 Scope

- 2.1 This report applies to the time interval indicated in the following table:

Table 2.1 – Stability Time Interval and Condition	
Time Interval	Condition
1 Year	Accelerated

3.0 References

- 3.1 TP-0057 - Stability Protocol, HDX Foley Catheter

4.0 Acceptance Criteria

4.1 Device Performance

4.1.1 Balloon Volume Maintenance

- 4.1.1.1 The catheter shall maintain its labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml) for 24-hour period.

4.1.2 Balloon Integrity (Resistance to Rupture)

- 4.1.2.1 The inflation balloon must be inflated easily with distilled or deionized water to its indicated labeled volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml) without showing any evidence of breakage throughout the 7-day test period.

4.1.3 Deflation Reliability (Failure to Deflate)

- 4.1.3.1 The balloon shall deflate to within 4 French sizes of the label French size within 15 minutes or be otherwise manipulated to effect drainage after the 7-day immersion period at its labeled fill volume (i.e., 14/18 Fr = 10ml, 16 Fr = 5ml).

4.1.4 Inflated Balloon Response to Traction

- 4.1.4.1 The entire balloon of catheters shall not pass into or through the funnel barrel and shall not rupture.

5.0 Deviations

- 5.1 There were no deviations to the evaluation protocol.

6.0 Results & Discussion

- 6.1 All test measurement equipment used in this evaluation was calibrated and documented on the appropriate data collection sheet.
- 6.2 The evaluation devices were EO sterilized. Refer to the Attachment 8.1 for additional details.

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6.3 The evaluation devices were placed into an environmental oven and subject to evaluated temperate of between 59-61°C from the period of July 12, 2019 to August 7, 2019, which twenty-six (26) days. This is equivalent to twelve (12) months. Refer to the Attachment 8.2 for the accelerated aging log.

6.4 Device Evaluation

6.4.1 Balloon Volume Maintenance

6.4.1.1 Twenty-two (22) units from each size was evaluated per test method indicated in TP-0057. The catheters were filled according to their labeled fill volumes (i.e., 14/18 Fr = 10ml, 16 Fr = 5 ml). All units evaluated maintained the delivered fill volume for the test duration. Acceptance criteria was met. Refer to attached data collection sheets for additional information.

6.4.2 Balloon Integrity (Resistance to Rupture)

6.4.2.1 Twenty-two (22) units from each size was evaluated per test method indicated in TP-0057. The catheters were filled according to their labeled fill volumes (i.e., 14/18 Fr = 10ml, 16 Fr = 5 ml). All units evaluated were not observed with leaks or breakage for the test duration. Acceptance criteria was met. Refer to attached data collection sheets for additional information.

6.4.3 Deflation Reliability (Failure to Deflate)

6.4.3.1 Twenty-two (22) units from each size was evaluated per test method indicated in TP-0057. The catheters were filled according to their labeled fill volumes (i.e., 14/18 Fr = 10ml, 16 Fr = 5 ml). Refer to attached data collection sheets for additional information.

6.4.3.2 The results for post-deflation measurements are indicated in the following table:

Table 6.5.3.2 – Statistical Summary, Deflation Reliability (inches)			
	14Fr (0.184)	16Fr (0.219)	18Fr (0.236)
Fill Volume	10ml	5ml	10ml
Min	0.206	0.226	0.246
Max	0.219	0.239	0.262
Mean	0.212	0.233	0.253
StDev	0.004	0.004	0.005
Spec	+ 4Fr (0.052)	+ 4Fr (0.052)	+ 4Fr (0.052)
Result	P	P	P

None of the catheters evaluated were observed with a rupture or leak and they all deflated with no issues. The post-deflation measurements were within the specified 4Fr from the labeled size. Acceptance criteria was met. Refer to attached data collection sheets for additional information.

6.4.4 Inflated Balloon Response to Traction

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- 6.4.4.1 Twenty-two (22) units from each size was evaluated per test method indicated in TP-0057. A fill volume of 5ml was evaluated since it is worst case for this evaluation. All units evaluated did not rupture or pass into the or through the funnel barrel in the static or impact load evaluation. Acceptance criteria was met. Refer to attached data collection sheets for additional information.

7.0 Conclusion

- 7.1 The HDX Foley Catheter met the acceptance criteria for device performance per §4.0 for product aged to one (1) year with accelerated conditioning.

8.0 Attachments

- 8.1 DHR-8000_A_LN_HYDPLX-14FR-ENG-003
- 8.2 DHR-8001_A_LN_HYDPLX-16FR-ENG-004
- 8.3 DHR-8002_A_LN_HYDPLX-18FR-ENG-005
- 8.4 STERIS Sterilization Process Record Load 21668
- 8.5 Accelerated Aging Log, T=1 Year @ 60°C (7/12/2019 – 8/7/2019)
- 8.6 Data Collection Sheet, Device Evaluation 14Fr (Dated: 9/5/2019)
- 8.7 Data Collection Sheet, Device Evaluation 16Fr (Dated: 9/5/2019)
- 8.8 Data Collection Sheet, Device Evaluation 18Fr (Dated: 9/5/2019)

9.0 Change History

Revision	Change Description
A	Initial release

Pathway Innovation LLC.	Design Failure Mode Effects Analysis (FMEA) – Risk Management Report HDX Foley Catheter	Doc No.: RM-0020
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1.0 Purpose

- 1.1 The purpose of this document is to identify the associated risk with the intended use of the HDX Foley Catheter as it relates to the design of the components and their interactions with each other.

2.0 Scope

- 2.1 The product numbers associated with the HDX Foley Catheter is indicated in the following table:

Table 2.1 – HDX Foley Catheter Products	
Part No.	Description
HDX-8000	14 Fr, 2-Way Foley Catheter
HDX-8001	16 Fr, 2-Way Foley Catheter
HDX-8002	18 Fr, 2-Way Foley Catheter

3.0 References

- 3.1 QP-004 – Risk Management

4.0 Risk Tables Definitions

- 4.1 Description – Describes the operation that is being evaluated for risk
- 4.2 Detection - The ability to screen out a hazard through verification activities (e.g., inspection).
- 4.2.1 Detection¹ (D¹) – The detection assessment is post the implementation/ verification of initial risk controls (i.e., Risk Controls¹)
- 4.2.2 Detection² (D²) – The detection assessment is post the implementation/ verification of the secondary risk controls (i.e., Risk Controls²)
- 4.3 Verification – Evidence source of linkage that initial (i.e., Risk Controls¹) risk controls have implemented, verified, or validated.
- 4.4 Probability – Likelihood of the hazard occurring
- 4.4.1 Probability¹ (P¹) – The probability assessment is post the implementation/ verification of initial risk controls (i.e., Risk Controls¹)
- 4.4.2 Probability² (P²) – The probability assessment is post the implementation/ verification of the secondary risk controls (i.e., Risk Controls²)
- 4.5 Potential Effect of Failure – All conceivable effects of the potential failure mode
- 4.6 Potential Cause of Failure – All conceivable causes of the potential failure mode
- 4.7 Potential Failure Mode (FM) – The potential consequences of the failure

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4.8 Risk Controls - Process through which decisions are reached and protective measures are implemented for reducing risk to, or maintaining risk within specified levels.

4.8.1 Risk Controls¹ – Initial risk controls that are incorporated.

4.8.2 Risk Controls² – Secondary risk controls, also known as “Mitigation”.

4.9 Risk Rating – Residual risk determination based on the concept of ‘As Far As Possible’ (AFAP) approach. The AFAP approach provides three (3) possible outcomes, which are 1) the residual risk exceeds the risk acceptability (RED), 2) the residual risk is acceptable to the risk acceptability criteria (GREEN), and 3) the residual risk is between the two (2) states indicated and it can be an option to lower the risk to the lowest practical level without economic considerations (YELLOW).

4.9.1 Risk Rating¹ (R¹) – This calculation is post the implementation of initial risk controls (i.e., Risk Controls¹)

4.9.2 Risk Rating² (R²) – This calculation is post the implementation of the secondary risk controls (i.e., Risk Controls²)

4.10 Severity (S) – Measure of the possible consequences of a hazard.

5.0 Risk Acceptability Determination

5.1 The criteria of risk acceptability will be determined with a Risk Priority Number (RPN) and on the concept of ‘As Far As Possible (AFAP)’ approach. Refer to the Risk Rating definition in §4.0.

5.2 There are three (3) components used in determining risk acceptability, which are Severity, Probability, and Detection. Refer to the following tables:

Table 5.2a - Severity Ratings	
Index	Description
1	No patient effect
2	Minor injury that resolves without treatment
3	Moderate injury that may require treatment
4	Serious injury requiring medical intervention
5	Life threatening injury, disability, death

Table 5.2b – Probability Ratings	
Index	Description
1	Improbable - <0.001%
2	Remote – 0.001% to .01%
3	Occasional – 0.01% to 0.1%
4	Likely – 0.1% to 1%
5	Frequent - >1%

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Table 6.0 – HDX Foley Catheter																
#	Process Description / Function	Potential Failure Mode	S	Potential Effects of Failure	Potential Causes of Failure	Risk Controls ¹	Verification	P ¹	D ¹	F ¹	R ¹	Risk Controls ²	P ²	D ²	F ²	R ²
6	Information for Use (DFU) – Provides device information	Complication with device use/ intended purpose	3	Patient harm	Incomplete or ambiguous instructions for use	IFU will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	1	1	3	N/A	N/A	N/A	N/A	N/A
7	Inner Carton – Protects and orients the pouched devices	Packaging is damaged	3	Device/ sterile barrier damage	Inadequate design	Material/ card weight selection (Design)	Design verification (distribution evaluation) TR-0074	1	1	1	3	N/A	N/A	N/A	N/A	N/A
						Geometry is design to handle anticipate load (Design)										
8	Label, Inner Carton – Provides device information	Identification issues	1	Nuisance in preparation for the clinical application	Incomplete or ambiguous product information	Label and labeling will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	1	1	1	N/A	N/A	N/A	N/A	N/A
		Device use outside specification	3	Patient harm	Inadequate disclosure of specifications/ limitation (e.g., single-use, expiry)	Label and labeling will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	1	1	3	N/A	N/A	N/A	N/A	N/A
9	Shipper Case – Protects packaged	Packaging is damaged	3	Device/ sterile barrier damage	Inadequate design	Material/ card weight selection (Design)	Design verification (distribution	1	1	1	3	N/A	N/A	N/A	N/A	N/A

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Table 6.0 – HDX Foley Catheter																
#	Process Description / Function	Potential Failure Mode	S	Potential Effects of Failure	Potential Causes of Failure	Risk Controls ¹	Verification	P ¹	D ¹	F ¹	R ¹	Risk Controls ²	P ²	D ²	F ²	R ²
	devices					Geometry is design to handle anticipate load (Design)	evaluation) TR-0074									
10	Label, Shipper Case – Provides device information	Identification issues	1	Nuisance in preparation for the clinical application	Incomplete or ambiguous product information	Label and labeling will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	1	1	1	N/A	N/A	N/A	N/A	N/A

7.0 Risk/ Benefit Assessment

7.1 The following table indicates the risk/benefit analysis for the residual risks that were greater than the risk acceptability limit after all applicable controls/mitigations were applied:

Table 7.1 – Risk/Benefit Assessment for Residual Risk above Acceptability Limit for Process Hazard Assessment									
Table No.	Process Description	Section No.	Potential Failure Mode	Potential Effect of Failure	Potential Cause of Failure	S	F ²	Risk/Benefit Analysis	Does the benefit outweigh the residual risk?
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

7.2 There was no residual risk identified that was above the risk acceptability limit per §5.3.

8.0 Conclusion

8.1 Risks have been reduced as low as possible by the following methods:

- Design the product with inherent safety
- Protective measures/ mechanisms to the product of the device.
- Provide information for safety

Pathway Innovation LLC.	Design Failure Mode Effects Analysis (FMEA) – Risk Management Report	Doc No.: RM-0020
	HDX Foley Catheter	Rev.: A
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Based on a review of the risk-management process for the HDX Foley Catheter described herein, the risk management plan based on ISO 14971:2012 has been implemented and the individual, cumulative and overall residual risk have been reduced as low as possible regardless of the magnitude and the benefit of using this device will outweigh risks posed by the device during usage.

9.0 Change History

Revision	Change Description
A	Initial release

	Product Requirements Specification / Traceability Matrix Hydrophilix Foley Products			Doc No.	PS-0006
				Revision	B
General Information					
Attribute	Description				
Purpose	To document the design requirements for the Hydrophilix Foley Catheter products and provide traceability between the risk management reports (i.e, Hazard Analysis, Design Failure Mode Effects Analysis) and Verification & Validation deliverables.				
Scope	Hydrophilix Foley Catheter family of products				
Device Name	HYDROPHILIX 100% Silicone 2-Way Foley Catheter, 14Fr/10cc HYDROPHILIX 100% Silicone 2-Way Foley Catheter, 16Fr/5cc HYDROPHILIX 100% Silicone 2-Way Foley Catheter, 18Fr/10cc				
Device's Intended Use / Indications for Use Statement	Urological catheter is intended for drainage of the urinary tract. Catheterization is accomplished by inserting the catheter through the urethra and into the bladder. Intended population is adults and pediatrics.				
User Types	Healthcare practitioners / Clinical user (e.g. nurses, doctors)				
User Environment	Hospital				
Patient Population	Any patient population				
Region or Country	US				
Agency	FDA				
Device Class	II				
Regulation - Description	21 CFR 876.5130 - Urological catheter and accessories				
Product Code	EZL - Catheter, Retention Type, Balloon				

	Product Requirements Specification / Traceability Matrix Hydrophilix Foley Products						Doc No.	PS-0006
							Revision	B
Requirement & Trace Matrix								
1	2	3	4	5	6	7	8	9
Type	User/Customer Requirement	Product Requirement Description	Applicable Feature of Device	REQ No.	Regulations/ Standards Reference	Risk Management Reference	Design Verification & Validation Evidence	Specification Reference / Design Output
Biological	N/A	The device shall met biocompatability requirement for a surface device with mucosal membrane contact for a prolong exposure >24 to 30 days, which the following evaluations are required: 1. Cytotoxicity 2. Sensitization 3. Irritation or Intracutaneous Reactivity 4. Material Mediated Pyrogen	Device	1	1, 7, 9		TR-0071	N/A
Labeling	N/A	Information shall be presented in symbol if the symbol is recognized by the regulatory agency(s)	Labeling	2	4, 10, 14		N/A	Refer to product labeling (HDX-4XXX) indicated in BOM for the following finished products: HDX-8000 HDX-8001 HDX-8002
Labeling	N/A	Product labeling (including direction for use) shall comply to FDA regulations on labeling. Specific requirement for devices with antimicrobial agents is indicated in guidance document (14).	Labeling	3	8, 10, 13		N/A	Refer to product labeling (HDX-4XXX) indicated in BOM for the following finished products: HDX-8000 HDX-8001 HDX-8002

	Product Requirements Specification / Traceability Matrix Hydrophilix Foley Products						Doc No.	PS-0006
							Revision	B
	Requirement & Trace Matrix							
1	2	3	4	5	6	7	8	9
Type	User/Customer Requirement	Product Requirement Description	Applicable Feature of Device	REQ No.	Regulations/ Standards Reference	Risk Management Reference	Design Verification & Validation Evidence	Specification Reference / Design Output
Labeling	N/A	Product labeling shall include the following information associated with the device description: 1. Outer Diameter (French Size) 2. Balloon Size (cc) 3. Effective Shaft Size 4. Type of Coating (if applicable) 5. Amount/ concentration of antimicrobial agent (if applicable)	Labeling	4	7, 13		N/A	Refer to product labeling (HDX-4XXX) indicated in BOM for the following finished products: HDX-8000 HDX-8001 HDX-8002
Material	N/A	Connector shall be made of rigid or semi-rigid materials.	Connector	5	12		N/A	HDX-6000 HDX-6001 HDX-6002
Material	N/A	The device shall be constructed on medical grade silicone material.	Device	6	N/A		N/A	HDX-6000 HDX-6001 HDX-6002
Packaging	N/A	The device will be packaged in a tyvek/poly pouch., which 10 pouched devices will be packaged in an box.	Packaged Device	7	N/A		N/A	HDX-8000 HDX-8001 HDX-8002
Packaging	N/A	The peel strength for 1" MFG sealed section of the sterile barrer packaging shall be ≥ 1 lb.	Sterile Barrier	8	3		TR-0074	N/A
Packaging	N/A	The sterile barrier packaging (include sealed area) shall not exhibit the following visual defects: 1. Unsealed Areas 2. Nonhomegeneous or Undersealed Areas 3. Oversealed Areas 4. Narrow Seals 5. Channels 6. Wrinkles/Foldovers/Cracks 7. Tears/Pinholes	Sterile Barrier	9	3		TR-0074	N/A
Packaging	N/A	The sterile barrier packaging shall not exhibit a constant stream of bubbles when submerge in water and internal pressurized to a determine test pressure.	Sterile Barrier	10	3		TR-0074	N/A

	Product Requirements Specification / Traceability Matrix Hydrophilix Foley Products						Doc No.	PS-0006
							Revision	B
Requirement & Trace Matrix								
1	2	3	4	5	6	7	8	9
Type	User/Customer Requirement	Product Requirement Description	Applicable Feature of Device	REQ No.	Regulations/ Standards Reference	Risk Management Reference	Design Verification & Validation Evidence	Specification Reference / Design Output
Performance	N/A	Balloon Integrity (Resistance to Rupture) - The balloon shall be inflated easily with distilled or deionized water to labeled volume without showing any evidence of breakage throughout a 24 hour period.	Device	11	7		TR-0072	N/A
Performance	N/A	Balloon Volume Maintenance - The balloon shall maintain its volume throughout a 24 hour period.	Device	12	7		TR-0072	N/A
Performance	N/A	Connector shall be non-interconnectables, which does not appear to provide a secure connection when forcefully assembled and shall easily disengage.	Connector	13	12		TR-0072	N/A
Performance	N/A	Deflation Reliability (Failure to Deflate) - The balloon shall deflate to within 4 French sizes of the label French size within 15 min or be otherwise manipulated to effect drainage within this time period.	Device	14	7		TR-0072	N/A
Performance	N/A	Flow Rate though Drainage Lumen - Shall have a minimum average flow rate of 100 cm3/min.	Device	15	7		TR-0072	N/A
Performance	N/A	Inflated Balloon Response to Traction - The entire, inflated balloon of catheters shall not pass into or through the funnel barrel neck (28 Fr x 6 cm) when a 1 lb weight is affixed from the distal catheter end and subjected to static (2 minutes) and dynamic load (2 ft drop).	Device	16	7		TR-0072	N/A
Performance	N/A	The device shall be easily inserted through the urethra until the deflated balloon passes through the urethral sphincter and in a manner that prevents urethral trauma.	Device	17	N/A		TR-0072	N/A

	Product Requirements Specification / Traceability Matrix						Doc No.	PS-0006
	Hydrophilix Foley Products						Revision	B
Requirement & Trace Matrix								
1	2	3	4	5	6	7	8	9
Type	User/Customer Requirement	Product Requirement Description	Applicable Feature of Device	REQ No.	Regulations/ Standards Reference	Risk Management Reference	Design Verification & Validation Evidence	Specification Reference / Design Output
Physical	N/A	Balloon & Shaft Size - The proximal tip, the balloon, and the shaft (for 10 in. distal to the balloon) shall meet the requirements on size and tolerances as is relates to the labeled French size per the tolerances ndicated below: Tip = ± 1 Fr Shaft = ± 1 Fr Balloon = + 4 Fr (Uninflated and Deflated after Immersion)	Device	18	7		TR-0072	HDX-6000 HDX-6001 HDX-6002
Physical	N/A	The balloon size shall be 10cc.	Device	19	N/A		TR-0072	HDX-6000 HDX-6001 HDX-6002
Physical	N/A	The catheter shall have the ability to flush the outer surface proximal to the inflated balloon.	Device	20	N/A		TR-0072	HDX-6000 HDX-6001 HDX-6002
Physical	N/A	the device shall be able to connect into an industry standard or equivalent urine bag to facilitate sterile and easy urine collection.	Device	21	N/A		TR-0072	HDX-6000 HDX-6001 HDX-6002
Physical	N/A	The device shall be equipment with 3 lumens, which have the following purposes: 1. Drainage 2. Balloon Inflation 3. Irrigation	Device	22	N/A		N/A	HDX-6000 HDX-6001 HDX-6002
Physical	N/A	The product offering will be French Sizes (Outer Diameter) of 14 Fr (0.184"), 16 Fr (0.210"), and 18 Fr (0.236").	Device	23	N/A		N/A	HDX-6000 HDX-6001 HDX-6002
Stability	N/A	The device shall have a final shelf-life period of 3 years, but initially be qualified for shorter shelf-life periods (e.g., 1 year).	Sterile, Packaged Device	24	N/A		TR-0073	N/A
Sterilization	N/A	Device shall be sterile to a sterility assurance level (SAL) of 10-6 when terminally sterilized via Ethlyene Oxide (EO).	Device	25	2		TR-0075	N/A
Performance	N/A	Packaged devices shall maintain it sterility and functionality through transportation	Sterile, Packaged Device	26	5, 6		TR-0074	N/A
User	N/A	A healthcare provider must be able to easily and quickly implement device.	Device	27	N/A		N/A	HDX-4001

	Product Requirements Specification / Traceability Matrix Hydrophilix Foley Products			Doc No.	PS-0006
				Revision	B
	Reference Documents				
Source No	Organization	Document No	Edition	Title	
1	ANSI/ AAMI/ ISO	10993-1	2009	Biological evaluation of medical devices - Part 1: Evaluation and testing within a risk management process	
2	ANSI/ AAMI/ ISO	11135	2014	Sterilization of health care products – Ethylene Oxide – Requirements for development, validation and routine control of a sterilization	
3	ISO	11607-1	2019	Packaging for terminally sterilized medical devices – Part 1: Requirements for materials, sterile barrier systems and packaging	
4	ANSI/ AAMI/ ISO	15223-1	2016	Medical devices — Symbols to be used with medical device labels, labeling and information to be supplied	
5	ASTM	D4169	2016	Standard Practice For Performance Testing Of Shipping Containers And Systems	
6	ASTM	D4332	2014	Standard Practice for Conditioning Containers, Packages, or Packaging Components for Testing.	
7	ASTM	F623	2019	Standard Performance Specification for Foley Catheter	
8	FDA	21 CFR 801	4/15/2018	Title 21 - Food and Drugs Chapter I – Food and Drug Administration Department of Health Services; Subchapter H – Medical Devices; Part	
9	FDA	FDA-2013-D-0350	6/16/2016	FDA Guidance: Use of International Standard ISO 10993, “Biological Evaluation of Medical Devices Part 1: Evaluation and Testing within a	
10	FDA	N/A	1/21/2000	Alternative to Certain Prescription Device Labeling Requirements	
11	ISO	14971	2012	Medical Devices - Application of Risk Management to Medical Devices	
12	ISO	80369-1	2018	Small-Bore Connectors For Liquids And Gases In Healthcare Applications - Part 1: General Requirements.	
13	FDA	N/A	N/A	Guidance for the Content of Premarket Notifications for Conventional and Antimicrobial Foley Catheters	
14	FDA	81 FR 38911	6/15/2016	Use of Symbols in Labeling	

	Product Requirements Specification / Traceability Matrix	Doc No.	PS-0006
	Hydrophilix Foley Products	Revision	B
Revision History			
Rev	Description of Changes		
A	Initial Release		
B	Update format, indications for use		

	Design Development Plan, Hydrophilix Foley Catheter		Doc No.: PL-0013
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1.0 Purpose

- 1.1 The purpose of this document is to provide details associated with design and development project for initial market release of the Hydrophilix (HDX) Foley Catheter. This project is documented under PRJ-0017.

2.0 Scope

- 2.1 The plan applies to the HDX Foley Catheter described below, which will be initial distributed in the US.

3.0 References

- 3.1 PRJ-0017 – Design Project, Hydrophilix Foley Catheter Product Introduction (US)
- 3.2 QP-007– Design Control

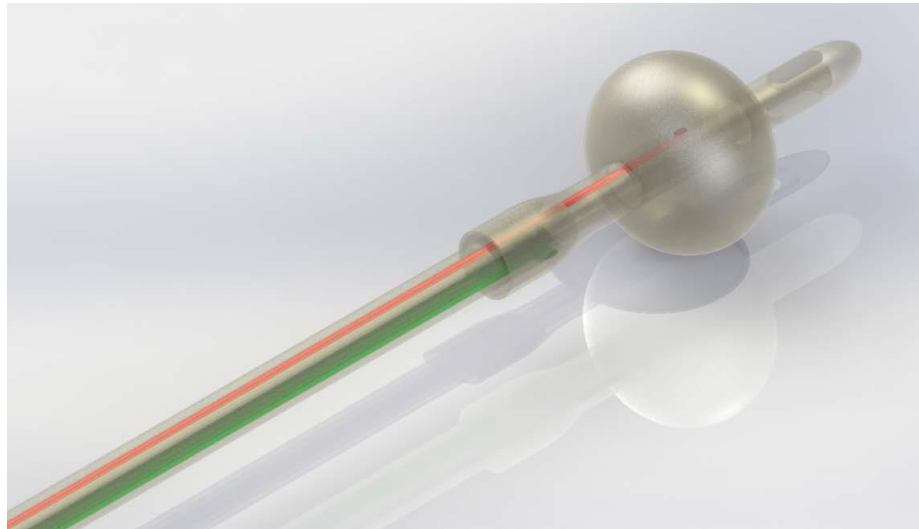
4.0 Background

- 4.1 Hydrophilix, Inc. was spun-out of the University of California, Los Angeles based on the Hydrophilix coating technology, a direct result of federal, state, and local support. The proof-of-concept began with basic research funding from the National Science Foundation Sustainable Chemistry Program and fellowships for individual scholars. The technology was further developed for commercial application, propelled with seed funding from the NSF Clean Energy for Green Industry, UCLA's Physical Sciences Entrepreneurship and Innovation Fund, and private investment.
- 4.2 When devices are processed with the Hydrophilix solution, a thin hydration layer prevents pathogens from adhering to the device surface and proliferating into biofilms. Because the processed surface does not kill microbes, the creation of superbugs is avoided. The processed surface is non-toxic and is applied in minutes.

5.0 Product Description

- 5.1 The HDX Foley catheter is a single use device that intended for use in the drainage and/or collection and/or measurement of urine. It is constructed of 2 lumens, which are for balloon inflation and drainage. The catheter will be constructed of medical grade silicone. The size offering will be 14, 16, and 18Fr with balloon volumes of 5ml – 30 ml. Refer to the following illustration of the device:

	Design Development Plan, Hydrophilix Foley Catheter		Doc No.: PL-0013
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6.0 Product Configuration

- 6.1 The product will be offered in 3 sizes initially: 14Fr 10ml, 16Fr 5ml, and 18Fr 10ml. Additional sizes will be added as they are available.
- 6.2 Devices will be packaged individually in a sterile barrier Tyvek/poly pouch, which 10 units will be incorporated per box.

7.0 Indication for Use Statement

- 7.1 Urological catheter intended for drainage/irrigation of urinary tract. Catheterization is accomplished by inserting the catheter through the urethra and into the bladder. Intended population is adults and pediatrics.

8.0 Goals & Objectives

- 8.1 Complete the following Design Control Activities defined in QP-007
 - 8.1.1 Planning
 - 8.1.2 Design Input
 - 8.1.3 Design Output
 - 8.1.4 Design Verification and Validation
 - 8.1.5 Prepare and submit FDA pre-market notification (510k) application and gain clearance to distribute in the US.
 - 8.1.6 Transfer the product to manufacturing to support commercialization of product. Note: This phase will be split into the following two (2) subphases.
 - 8.1.6.1 Manual process
 - 8.1.6.2 Automated process

9.0 Resources

	Design Development Plan, Hydrophilix Foley Catheter		Doc No.: PL-0013
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- 9.1 Refer to the following table for the internal and external resources that are required to execute the design and development plan:

Internal/ External	Function	Name	Title
External	Customer	Brian McVerry	CTO
Internal	R&D	Arthur Deptala	Managing Partner
Internal	Manufacturing / Project Leader	Jonas Cochran	Managing Partner
Internal	Regulatory/ Quality	Aaron Rogers	Director of Regulatory and Quality
Internal	Independent Reviewer	Tom Cassou	Director of Business Development

- 9.2 External testing services are provided by:

9.2.1 Biocompatibility: Nelson Labs, WUXI

9.2.2 Sterilization: Steris

9.2.3 Transportation Testing: Westpak

- 9.3 Key Component and Process Suppliers:

9.3.1 Non-Sterile Foley Catheter: Ningbo Yingmed

9.3.2 Surface Treatment Solution: Hydrophilix, Inc.

9.3.3 Sterile Barrier Packaging: Steripax

9.3.4 Sterilization: Steris

10.0 Schedule/ Timelines

- 10.1 Schedules and timelines for each phase will be updated throughout the project and is maintained as part of the project module in Arena PLM.

11.0 Change History

Revision	Change Description
A	Initial Release
B	Update resources §9.0, Indications for Use §7.0, project phases §8.0, product configuration §6.0

	Regulatory Plan, Hydrophilix Foley Catheter		Doc No.: PL-0014
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1.0 Purpose

- 1.1 The purpose of this document is to provide the details of a regulatory plan for initial market release of the Hydrophilix (HDX) Foley Catheter. This project is documented under PRJ-00017.

2.0 Scope

- 2.1 The plan applies to the applies to the HDX Foley Catheter described below, which will be initial distributed in the US.

3.0 References

- 3.1 PRJ-0017 – Design Project, Hydrophilix Foley Catheter Product Introduction (US)
- 3.2 QP-007– Design Control

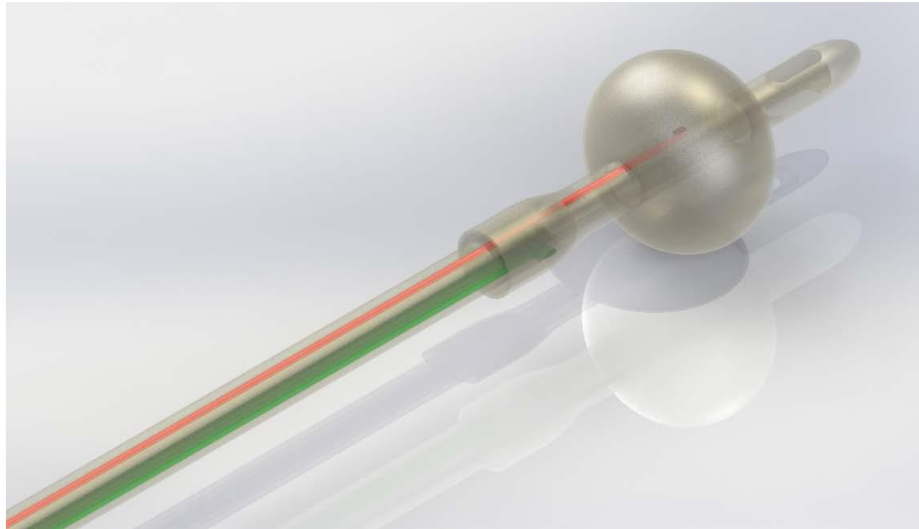
4.0 Background

- 4.1 Hydrophilix, Inc. was spun-out of the University of California, Los Angeles based on the Hydrophilix coating technology, a direct result of federal, state, and local support. The proof-of-concept began with basic research funding from the National Science Foundation Sustainable Chemistry Program and fellowships for individual scholars. The technology was further developed for commercial application, propelled with seed funding from the NSF Clean Energy for Green Industry, UCLA's Physical Sciences Entrepreneurship and Innovation Fund, and private investment.
- 4.2 When devices are processed with the Hydrophilix solution, a thin hydration layer prevents pathogens from adhering to the device surface and proliferating into biofilms. Because the processed surface does not kill microbes, the creation of superbugs is avoided. The processed surface is non-toxic and is applied in minutes.

5.0 Product Description

- 5.1 The HDX Foley catheter is a single use device that intended for use in the drainage and/or collection and/or measurement of urine. It is constructed of 2 lumens, which are for balloon inflation and drainage. The catheter will be constructed of medical grade silicone. The size offering will be 14, 16, and 18Fr with balloon volumes of 5ml – 30 ml. Refer to the following illustration of the device:

	Regulatory Plan, Hydrophilix Foley Catheter		Doc No.: PL-0014
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6.0 Commercialization/ Distribution Strategy

6.1 US Market

7.0 Regulatory Strategy

7.1 US Market

7.1.1 Device Classification: Class II

7.1.1.1 Product Code: EZL

7.1.1.2 Regulation Number: 876.5130

7.1.2 Market Entry Requirements: Premarket Notification (510k)

7.1.3 Predicate Devices

7.1.3.1 K091767 – Cook 3 Way Silicone Foley Balloon Catheter

7.1.3.2 K142636 – Medline Silicone Foley Catheter

7.1.3.3 K181616 – PSM Silicone Foley Catheter

7.1.4 Facility Registration & Device Listing: Establishment Activities

7.1.4.1 Contract Manufacturer: Pathway LLC (Establishment Registration No. 3012833053)

7.1.4.2 Specification Developer & Complaint Handler: Hydrophilix

8.0 Clinical Considerations

8.1 Indication for Use

8.1.1 Urological catheter intended for drainage/irrigation of urinary tract. Catheterization is accomplished by inserting the catheter through the urethra and into the bladder. Intended population is adults and pediatrics.

8.2 User Types

8.2.1 Pediatrics & Adults, Male & Female

8.3 User Environment

	Regulatory Plan, Hydrophilix Foley Catheter		Doc No.: PL-0014
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8.3.1 Healthcare Facility, Prescription Use

9.0 Associated Product Claims

- 9.1 There are no claims outside of the indication for use statement referenced in §8.1

10.0 Shelf-Life Considerations

- 10.1.1 A minimum shelf-life of one (1) year with the end goal of three (3) years.

11.0 Applicable Guidance Documents, Standards, and Regulations

11.1 Market Clearance

- 11.1.1 21 CFR 807 – Establishment registration and device listing for manufacturers and initial importers of devices

11.1.1.1 Subpart B – Procedure for Device Establishments

11.1.1.2 Subpart E – Premarket Notification Establishment

- 11.1.2 Guidance for the Content of Premarket Notifications for Conventional and Antimicrobial Foley Catheters

11.2 Quality Management System

- 11.2.1 21 CFR 820 – Quality System Regulations

- 11.2.2 21 CFR 803 – Medical Device Reporting

- 11.2.3 21 CFR 806 – Medical Devices; Reports of Corrections and Removals

11.3 Risk Management

- 11.3.1 ISO 14971:2012 – Medical Devices - Applications Of Risk Management To Medical Devices

11.4 Labeling

- 11.4.1 21 CFR 801 – Labeling

- 11.4.2 Guidance for Industry – Alternative to Certain Prescription Device Labeling Requirements – 1/21/2000

- 11.4.3 81 FR 38911 – Use of Symbols on Labeling

- 11.4.4 ANSI / AMII / ISO 15223-1:2016 – Medical devices — Symbols to be used with medical device labels, labeling and information to be supplied.

11.5 Biosafety

- 11.5.1 ANSI / AAMI / ISO 10993-1:2009 – Biological Evaluation of Medical Devices - Part 1: Evaluation and Testing Within a Risk Management Process

- 11.5.2 FDA-2013-D-0350, June 16, 2016 – FDA Guidance: Use of International Standard ISO 10993, “Biological Evaluation of Medical Devices Part 1: Evaluation and Testing within a risk management process”

11.6 Performance

- 11.6.1 ASTM F623: 2019 – Standard Performance Specification for Foley Catheter

- 11.6.2 ISO 80369-1: 2018 – Small-Bore Connectors For Liquids And Gases In Healthcare Applications - Part 1: General Requirements

	Regulatory Plan, Hydrophilix Foley Catheter		Doc No.: PL-0014
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11.7 Stability

11.7.1 ASTM F1980-16 – Standard Guide for Accelerated Aging of Sterile Barrier Systems for Medical Devices

11.8 Sterilization

11.8.1 ANSI / AAMI / ISO 11135:2014 - Sterilization of health care products – Ethylene Oxide – Requirements for development, validation and routine control of a sterilization process for medical devices

11.9 Packaging

11.9.1 ISO 11607-1:2019 – Packaging for Terminally Sterilized Medical Devices - Part 1: Requirements for Materials, Sterile Barrier Systems and Packaging Systems

11.9.2 ASTM D4332-14 – Standard Practice for Conditioning Containers, Packages, Or Packaging Components for Testing

11.9.3 ASTM D4169-16 – Standard Practice for Performance Testing of Shipping Containers and Systems.

12.0 Change History

Revision	Change Description
A	Initial Release
B	Update reg & listing §7.1.4, indications for use §8.1.1

Pathway Innovation LLC.	Risk Management Plan – Hydrophilix Foley Catheter	Doc No.: RM-0018
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		Page 1 of 2

1.0 Purpose

The purpose of this document is to indicate the requirements of the risk management plan that is associated with the development of the Hydrophilix (HDX) Foley Catheter.

2.0 Scope

2.1 The document applies for the design project associated with the product introduction of the HDX Foley Catheter.

2.2 This document will identify the activities planned for risk management.

3.0 References

3.1 ISO 14971:2012 – Medical devices – Application of risk management to medical devices

3.2 QP-002 – Document Control

3.3 QP-004 – Risk Management

3.4 QP-007 – Design Control

3.5 QP-011 – Post Market Surveillance

3.6 Project#00017 – Hydrophilix Foley Catheter product introduction (US)

4.0 Responsibilities & Authorities

4.1 The risk management team will include members of Pathway. Team members will include individuals representing Research & Development (R&D), Manufacturing, Marketing, Regulatory Affairs (RA), and Quality Assurance (QA).

Individuals that participate in risk management activities will have knowledge of and experience with the device (or similar device) and how it is manufactured.

4.2 The risk management team will be required to complete all activities indicated in this document. Team members from Pathway will be required to review all content of risk management deliverables and agree on its conclusions with documented approval.

5.0 Product Description

The HDX Foley Catheter is intended for use to drain/irrigate the urinary tract.

6.0 Product Life-Cycle Phases

6.1 Design

In the appropriate phase of the design control process, per QP-007, a Hazard Analysis (HA) and Design Failure Mode Effects Analysis (DFMEA) will be developed per, QP-004, as input to the design of the device.

Criteria of risk acceptability is identified in QP-004.

Pathway Innovation LLC.	Risk Management Plan – Hydrophilix Foley Catheter	Doc No.: RM-0018
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If additional risk(s) are identified during the progression through the design control phases, especially in verification and validations, risk management deliverables will be reviewed and updates appropriately.

6.2 Process Validation

In the “Transfer to Manufacturing” phase of the design control process per QP-007, a Process Failure Mode Effects Analysis (PFMEA) will be developed as a deliverable process validation. Criteria of risk acceptability is identified in QP-004.

6.3 Production & Post-Production

Information collected from production and post-production activities will be evaluated for risk as part of the post-market surveillance process per QP-011. Risk management report will be review and newly identified risk or changes to existing risk will be incorporated.

Revised information to the risk management reports will be reviewed with R&D, RA, and QA to determine if action shall be taken to the product design to mitigate identified risk. Action shall be incorporated in a change control project per QP-007.

7.0 Verification Activities

Activities associated for risk controls (initial & mitigation) will be documented in the risk management reports.

8.0 Risk Management File

The risk management file consists of this plan and the risk management report (i.e., Hazard Analysis, DFMEA, & PFMEA). The risk management plan will be maintained as a controlled document per QP-002 and associated in the product’s Design History File (DHF).

9.0 Change History

Revision	Change Description
A	Initial release

Pathway Innovation LLC.	Hazard Analysis (HA) – Risk Management Report, HDX Foley Catheter	Doc No.: RM-0019
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1.0 Purpose

- 1.1 The purpose of this document is to identify the associated risk with the intended use of the HDX Foley Catheter.

2.0 Scope

- 2.1 The product numbers associated with the HDX Foley Catheter is indicated in the following table:

Table 2.1 – HDX Foley Catheter Products	
Part No.	Description
HDX-8000	14 Fr, 2-Way Foley Catheter
HDX-8001	16 Fr, 2-Way Foley Catheter
HDX-8002	18 Fr, 2-Way Foley Catheter

- 2.2 The intended use of this device is the following:

Single-use urinary catheter that intended for use in the drainage / irrigation of the urinary tract.

3.0 References

- 3.1 QP-004 – Risk Management

4.0 Risk Tables Definitions

- 4.1 Description – Describes the operation that is being evaluated for risk
- 4.2 Verification – Evidence source of linkage that initial (i.e., Risk Controls¹) risk controls have implemented, verified, or validated.
- 4.3 Probability – Likelihood of the hazard occurring
- 4.3.1 Probability¹ (P¹) – The probability assessment is post the implementation/ verification of initial risk controls (i.e., Risk Controls¹)
- 4.3.2 Probability² (P²) – The probability assessment is post the implementation/ verification of the secondary risk controls (i.e., Risk Controls²)
- 4.4 Potential Hazard (PH) – Source of harm
- 4.5 Potential Failure Mode (FM) – The potential consequences of the failure
- 4.6 Risk Controls - Process through which decisions are reached and protective measures are implemented for reducing risk to, or maintaining risk within specified levels.
- 4.6.1 Risk Controls¹ – Initial risk controls that are incorporated.
- 4.6.2 Risk Controls² – Secondary risk controls, also known as “Mitigation”.

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4.7 Risk Rating – Residual risk determination based on the concept of ‘As Far As Possible (AFAP)’ approach. The AFAP approach provides three (3) possible outcomes, which are 1) the residual risk exceeds the risk acceptability (RED), 2) the residual risk is acceptable to the risk acceptability criteria (GREEN), and 3) the residual risk is between the two (2) states indicated and additional controls are suggested to lower the risk to the lowest practical level, without economically considerations, based on management decision (YELLOW).

4.7.1 Risk Rating¹ (R¹) – This calculation is post the implementation of initial risk controls (i.e., Risk Controls¹)

4.7.2 Risk Rating² (R²) – This calculation is post the implementation of the secondary risk controls (i.e., Risk Controls²)

4.8 Severity (S) – Measure of the possible consequences of a hazard.

5.0 Identify Characteristics that could Impact on Safety (Annex C)

C.2.1 What is the intended use and how is the medical device to be used? *Refer to §2.2.*

C.2.2 Is the medical device intended to be implanted? *No*

C.2.3 Is the medical device intended to be in contact with the patient or other persons? *Yes*

C.2.4 What materials or components are utilized in the medical device or are used with, or are in contact with, the medical device? *Silicone*

C.2.5 Is energy delivered to or extracted from the patient? *No*

C.2.6 Are substances delivered to or extracted from the patient? *Yes, substances can be delivery to the urinary tract.*

C.2.7 Are biological materials processed by the medical device for subsequent re-use, transfusion or transplantation? *No*

C.2.8 Is the medical device supplied sterile or intended to be sterilized by the user, or are other microbiological controls applicable? *Yes*

C.2.9 Is the medical device intended to be routinely cleaned and disinfected by the user? *No*

C.2.10 Is the medical device intended to modify the patient environment? *No*

C.2.11 Are measurements taken? *Yes, measurements of urine.*

C.2.12 Is the medical device interpretative? *No*

C.2.13 Is the medical device intended for use in conjunction with other medical devices, medicines or other medical technologies? *Yes, urine collection bag.*

C.2.14 Are there unwanted outputs of energy or substances? *No*

C.2.15 Is the medical device susceptible to environmental influences? *No*

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- C.2.16 Does the medical device influence the environment? *No*
- C.2.17 Are there essential consumables or accessories associated with the medical device? *No*
- C.2.18 Is maintenance or calibration necessary? *No*
- C.2.19 Does the medical device contain software? *No*
- C.2.20 Does the medical device have a restricted shelf-life? *Yes*
- C.2.21 Are there any delayed or long-term use effects? *No*
- C.2.22 To what mechanical forces will the medical device be subjected? *Tensile force from the urine content in the collection bag; Internal pressure from urine drainage or irrigation of the urinary tract.*
- C.2.23 What determines the lifetime of the medical device? *Product and sterile barrier performance*
- C.2.24 Is the medical device intended for single use? *Yes*
- C.2.25 Is safe decommissioning or disposal of the medical device necessary? *Yes*
- C.2.26 Does installation or use of the medical device require special training or special skills? *No*
- C.2.27 How will information for safe use be provided? *Direction for Use*
- C.2.28 Will new manufacturing processes need to be established or introduced? *No*
- C.2.29 Is successful application of the medical device critically dependent on human factors such as the user interface? *No*
 - C.2.29.1 Can the user interface design features contribute to use error? *No*
 - C.2.29.2 Is the medical device used in an environment where distractions can cause use error? *No*
 - C.2.29.3 Does the medical device have connecting parts or accessories? *Yes, urine collection bag*
 - C.2.29.4 Does the medical device have a control interface? *No*
 - C.2.29.5 Does the medical device display information? *No*
 - C.2.29.6 Is the medical device controlled by a menu? *No*
 - C.2.29.7 Will the medical device be used by persons with special needs? *No*
 - C.2.29.8 Can the user interface be used to initiate user actions? *No*
- C.2.30 Does the medical device use an alarm system? *No*
- C.2.31 In what way(s) might the medical device be deliberately misused? *None*

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- C.2.32 Does the medical device hold data critical to patient care? *No*
- C.2.33 Is the medical device intended to be mobile or portable? *Yes*
- C.2.34 Does the use of the medical device depend on essential performance? *No*

6.0 Risk Acceptability Determination

- 6.1 The criteria of risk acceptability will be determined with a Risk Priority Number (RPN) and on the concept of 'As Far As Possible' (AFAP) approach. Refer to the Risk Rating definition in §4.0.
- 6.2 There are two (2) components used in determining risk acceptability, which are Severity and Probability. Refer to the following tables:

Table 5.2a - Severity Ratings	
Index	Description
1	No patient effect
2	Minor injury that resolves without treatment
3	Moderate injury that may require treatment
4	Serious injury requiring medical intervention
5	Life threatening injury, disability, death

Table 5.2b – Probability Ratings	
Index	Description
1	Improbable - <0.001%
2	Remote – 0.001% to .01%
3	Occasional – 0.01% to 0.1%
4	Likely – 0.1% to 1%
5	Frequent - >1%

- 6.3 Risk acceptability will be determined per the following risk index:

Table 5.3 – Risk Rating Table						
Severity	5	5	10	15	20	25
	4	4	8	12	16	20
	3	3	6	9	12	15
	2	2	4	6	8	10
	1	1	2	3	4	5
	1	1	2	3	4	5
Probability						

7.0 Risk Assessment Tables

Table 6.0 –Kurin Blood Collection Device

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#	Hazard Category	PH #	Potential Hazard	S	Potential Failure Mode	Risk Controls ¹	Verification	P ¹	R ¹	Risk Controls ²	P ²	R ²
1	Electromagnetic Energy	The medical device does not have any electromagnetic energy hazards.										
2	Radiation Energy	The medical device does not have any radiation energy hazards.										
3	Thermal Energy	The medical device does not have any thermal energy hazards.										
4	Mechanical Energy	The medical device does not have any mechanical energy hazards.										
5	Biological	5.1	Fever/ infection to patient	3	Device is not provided sterile/ insufficient sterilization process	Device will be terminally sterilized with validated method (Design)	Sterilization validation	1	3	N/A	N/A	N/A
					Device is reused	Warnings statement for single-use (Information)	IFU	2	6	N/A	N/A	N/A
					Inadequate sterile barrier system	Sterile barrier system will be designed for material that are commonly used for single-use medical device (Design)	Design verification - sterile barrier	1	3	N/A	N/A	N/A
						Warning will be incorporated for inspecting packaging prior to use as well not using the device if packaging is damaged (Information)	Stability	2	6	N/A	N/A	N/A
					Use beyond expiration date	Warning statement (Information)	IFU	2	6	N/A	N/A	N/A
		5.2	Fever/infection to clinician	3	Proper handling and disposal is not conducted during clinical application	Instruction and warning statement (Information)	IFU	2	6	N/A	N/A	N/A
6	Chemical	6.1	Allergic reaction	3	Materials (e.g., latex, rubber) that is known to cause allergic response	Material will be selected that do not cause allergic response (Design)	Design verification - biocompatibility	1	3	N/A	N/A	N/A
		6.2	False positives in the blood work evaluation – red blood cell damage	1	Material leach during clinical application	Material selection (Design)	Design verification - biocompatibility	1	1	N/A	N/A	N/A
					Excessive ethylene oxide residuals from sterilization process	Device will be released for distribution with acceptable residual levels (Design)	Sterilization validation	1	1	N/A	N/A	N/A

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Table 6.0 –Kurin Blood Collection Device												
#	Hazard Category	PH #	Potential Hazard	S	Potential Failure Mode	Risk Controls ¹	Verification	P ¹	R ¹	Risk Controls ²	P ²	R ²
		6.3	Fever	3	Contacting material causes pyrogenic response	Material selection (Design)	Design verification - biocompatibility (i.e., Material mediated pyrogenicity)	1	3	N/A	N/A	N/A
		6.4	Procedure complication – device inoperative	3	Material degrades over usable life	Material selection (Design) Sterilization method selection (Design)	Stability evaluation	1	3	N/A	N/A	N/A
7	Biocompatibility	7.1	Cell Damage	3	Patient contact materials are not biocompatibility	Material selection (Design)	Design verification - biocompatibility	1	3	N/A	N/A	N/A
8	Function (Operational)	8.1	Procedure complication – device inoperative	2	Balloon Burst – Material or dimensional issues (wall thickness)	Material selection (Design) Dimensional Assessment (Design)	Design verification - Balloon Integrity (Resistance to Rupture)	1	2	N/A	N/A	N/A
				1	Balloon deflates or does not inflates – Valve malfunctions or leaks (wall thickness issue)	Valve selection (Design) Dimensional Assessment (Design)	Design verification - Balloon Volume Maintenance	1	1	N/A	N/A	N/A
				1	Balloon does not inflate – Valve malfunctions	Valve selection (Design)	Design verification	1	1	N/A	N/A	N/A
				2	Catheter breaks under normal tensile stresses - Material or dimensional issues	Valve selection (Design) Dimensional Assessment (Design)	Design verification - Inflated Balloon Response to Traction	1	2	N/A	N/A	N/A
				1	Connection issues – Syringe (inflation, irrigation); Collection bag (drainage)	Connection will be standardized to what is incorporated with marketed catheters (Design)	Design verification	1	1	N/A	N/A	N/A
		8.2	Procedure complication – downgraded performance	2	Poor drainage - Insufficient holes and lumen diameter, Lumen collapses (Materials or dimensional issues)	Material selection (Design) Dimensional Assessment (Design)	Design verification - Flow Rate though Drainage Lumen	1	2	N/A	N/A	N/A
				2	Poor irrigation - Insufficient holes and lumen diameter, Lumen collapses (Materials or dimensional issues)	Material selection (Design) Dimensional Assessment (Design)	Design verification	1	2	N/A	N/A	N/A

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#	Hazard Category	PH #	Potential Hazard	S	Potential Failure Mode	Risk Controls ¹	Verification	P ¹	R ¹	Risk Controls ²	P ²	R ²
					Device used after expiration date or outside biocompatibility duration limits							
					Occurs during disposal process							
10	Labeling (Product Labeling)	10.1	Loss of traceability - Cannot identify the device or trace back to MFG	1	Incomplete or ambiguous instructions for use	Label and labeling will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	1	N/A	N/A	N/A
		10.2	Device used outside specification (expiry/ single-use	1	Incomplete or ambiguous instructions for use	Label and labeling will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	1	N/A	N/A	N/A
11	Operating Instructions (Product Labeling)	11.1	Procedural Complication /Patient discomfort	3	Incomplete or ambiguous instructions for use	IFU will be designed to requirements indicated in regulation for the markets intended to be distributed (Information)	Label content review	1	3	N/A	N/A	N/A
12	Environmental	12.1	Device/ Sterile Barrier Failure /Patient health is compromised	3	Stored outside the prescribe environmental parameters	Warning statement (Information)	IFU	2	6	N/A	N/A	N/A
					Packaging is not adequate to protect the device through transportation	Packaging will be design to protect the device and it sterile barrier through transportation stresses (Design)	Design Verification (Distribution Evaluation)	1	3	N/A	N/A	N/A

8.0 Risk/ Benefit Assessment

8.1 The following table indicates the risk/benefit analysis for the residual risks that were greater than the risk acceptability limit after all applicable controls/mitigations were applied:

Table 7.1 – Risk/Benefit Assessment for Residual Risk above Acceptability Limit for Process Hazard Assessment										
Table No.	Process Description	Section No.	Potential Failure Mode	Potential Effect of Failure	Potential Cause of Failure	S	F ²	Risk/Benefit Analysis		Does the benefit outweigh the residual risk?
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		N/A

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8.2 There was no residual risk identified that was above the risk acceptability limit per §5.3.

9.0 Conclusion

9.1 Risks have been reduced as low as possible by the following methods:

- Design
- Appropriate testing to ensure design criteria were met
- Protective measures (critical-to-quality)
- Manufacturing and quality control measures

9.2 Based on a review of the risk-management process for the HDX Foley Catheter described herein, the risk management plan based on ISO 14971:2012 has been implemented and the individual, cumulative and overall residual risk have been reduced as low as possible regardless of the magnitude and the benefit of using this device will outweigh risks posed by the device during usage.

10.0 Change History

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